

Twinkl: An Inner Compass for Longitudinal Alignment Between Behaviour and Core Values

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Abstract

AI-assisted reflection can support self-examination, but a response that defaults to agreement may reinforce a user’s account without testing it against the available evidence. Twinkl investigates a narrower role: evidence-grounded accountability. It compares longitudinal Journal Entries with a confirmed Profile of Core Values, cites the reported behaviour behind each conclusion, and returns Insufficient Evidence when a failed review prevents a current Drift claim or when an Abstain decision or Journal Entry gap blocks such a claim after recent Conflict evidence. Schwartz’s Theory of Basic Human Values supplies an established theoretical foundation and a vocabulary of competing priorities.

Three linked empirical investigations organise the research. First, the project constructed 1,651 synthetic longitudinal Journal Entries from 204 personas under demographic, narrative-continuity, label-leakage, and train-only augmentation controls. A shared 115-Journal-Entry benchmark produced human-human Fleiss’ kappa of 0.56 and mean LLM-Judge-human Cohen’s kappa of 0.66, while two targeted synthetic batches produced local but not general improvements. Second, 69 run IDs covering 133 persisted configurations established a measurable but inadequate performance frontier for the Value Identity Function (VIF) Critic (Offline): the three-seed Balanced Softmax reference reached median quadratic weighted kappa (QWK) of 0.362 and Conflict recall of 0.313. Third, VIF Critic Predictions did not improve Drift recall in the tested reviewer-input or scheduling experiments, whereas the Weekly Drift Reviewer produced a more useful but still development-only operating point when it examined cumulative Journal Entry history. The entry-level VIF Critic (Offline) and longitudinal Weekly Drift Reviewer use different inputs and reference labels, so their scores are not a direct model comparison.

The implemented core assessment path combines Weekly Drift Reviewer Decisions with the deterministic Drift Detector. Automated tests verify evidence preservation, state transitions, and recovery from failed model calls, while Inspect exposes the prompts, recorded responses, and validations behind each result. A subsequent prompt study examined how Core Value definitions affected Drift detection, and a 501-week North Star Moment comparison found higher Card precision and Opportunity recall for full eligible history than Nomic top-three retrieval. Together, these studies and application checks support an integrated and inspectable capstone implementation on AI-reviewed synthetic development evidence. Real-user usefulness, future human calibration of the AI review, fresh final-test performance, and deployment approval remain open.

1. Introduction

1.1 Problem and objective

Personal informatics applications help people collect and reflect on personal data [1], [2], yet a record alone may not prompt reflection. Reflective informatics suggests that reflection can begin when an application helps a person notice a breakdown, ask a question, and reconsider an earlier assumption [3]. AI assistance introduces a further risk: models trained with human feedback can favour responses that match a user's stated beliefs over responses that challenge them [4]. In a reflective setting, this tendency creates a risk of ungrounded affirmation.

Twinkl investigates whether an AI-assisted reflection application can take a more accountable role. It maintains a confirmed Profile of Core Values, compares later Journal Entries with those Core Values over time, and requires cited behavioural evidence before declaring Drift. Here, Conflict means a Journal Entry that clearly describes behaviour against a value, and Drift requires two consecutive Conflicts for the same user-confirmed Core Value. Insufficient Evidence is a valid result, and Coach Digest asks a non-prescriptive question instead of rewarding or condemning the user. Twinkl does not measure or claim to eliminate sycophancy as a model property. It addresses the narrower product-level risk of ungrounded affirmation through evidence requirements, abstention, and an explicit sequence rule.

Schwartz's Theory of Basic Human Values supplies the vocabulary for Core Values. The original theory identified ten broad values and their motivational relations, and later work refined these into 19 narrower values while preserving the circular motivational continuum [5], [6]. Best-Worst Scaling adds a research-grounded way to elicit relative importance [7], [8]. This foundation gives the Profile academic provenance without turning it into a diagnosis or claiming that the Twinkl assessment is psychometrically validated.

The product objective is to give the user an evidence-grounded weekly reflection. This reflection depends on four design commitments: identifying specific Conflict evidence, applying the stable Drift Detector, citing the relevant Journal Entries, and using non-prescriptive language. A displayed nudge can ask one contextual question after an eligible Journal Entry. This immediate interaction is separate from Weekly Drift Detection.

Twinkl is not a clinical or therapeutic application. Research on conversational agents shows promise but also shows limited evidence, mixed study quality, and unclear benefit outside tested settings [9]. The application therefore uses explicit assessment-only language, model validation that withholds invalid output, evidence citations, and a separate Inspect view.

1.2 Intended users, practical context, and success criteria

Twinkl is intended for knowledge workers in transition, including graduate students, new managers, founders, and other professionals whose work, family, and personal-growth priorities compete across a week. The practical problem is not an inability to name what matters. The project hypothesis is that repeated choices can accumulate without an accountable comparison with the person's declared priorities and that episodic journaling alone may leave the pattern unnoticed.

Requirement elicitation was project-led rather than sponsor- or customer-led. The April proposal, the personal-informatics and values literature, synthetic Persona journeys, and iterative technical studies established four application requirements: construct a user-confirmed Profile, preserve chronological Journal Entry evidence, apply a contestable Drift rule, and present the result without prescription or hidden

authority. These requirements were refined when the VIF hand-off evidence changed which component received user-facing Drift authority.

The NUS-ISS capstone brief notes that student-proposed projects benefit from credible potential users who can assist with requirements, testing, and acceptance [10]. Twinkl did not complete this external-user step during development. The planned pilot must therefore test the requirements and acceptance criteria themselves, rather than merely ask users to rate a finished design.

Application success is therefore separated into implemented and user-validated criteria. The capstone application must complete an end-to-end walkthrough, preserve session and chronological state, expose the evidence and model receipts (saved records of model calls) behind each result, fail closed when a valid decision is unavailable, and confirm deletion of matching browser and temporary Python state. Perceived accuracy, relevance, timing, continued journaling, and customer satisfaction require the planned five-to-ten-user pilot and are not inferred from synthetic Persona replays.

1.3 Research objective and empirical investigations

The research objective is to evaluate a staged path from synthetically generated longitudinal Journal Entries to evidence-grounded Weekly Drift Detection, establishing what the VIF Critic (Offline) can learn and what an LLM-based Weekly Drift Reviewer contributes when the tested compact model is insufficient. The following three investigations retrospectively organise the project’s iterative R&D path; they are not presented as preregistered hypotheses.

Investigation 1 — constructing and validating a synthetic longitudinal corpus: What design and validation controls make Twinkl’s synthetic Journal Entries and LLM-Judge VIF Labels suitable for bounded model development, and what limitations remain?

Investigation 2 — testing the limits of a compact VIF Critic (Offline): What value-alignment signal can a compact VIF Critic (Offline) recover from the synthetic corpus, where does its performance break down across Schwartz values, and what do those limits imply for user-facing Drift authority?

Investigation 3 — detecting longitudinal Drift with an LLM-based Weekly Drift Reviewer: Do VIF Critic Predictions improve Weekly Drift Detection when supplied to the Weekly Drift Reviewer or used to trigger earlier review, and which model and reasoning-effort choices provide the best supported trade-off among Drift recall, false Drift alerts, and coverage?

Two subsequent studies examine how Core Value definitions affect Weekly Drift Detection and whether North Star Moment quotation selection benefits from full eligible history rather than retrieved candidates. Coach Digest generation and AI editorial review provide additional evidence about source interpretation and reflective language.

The implementation objective is to deliver one end-to-end React application in which an assessor can trace Profile construction, Weekly Drift Detection, and Coach Digest to recorded inputs, model receipts, validations, and deterministic state transitions. Verification tests the integrity of this path, including invalid provider responses, persistence, retries, and session deletion. Saved replays and automated tests establish the exercised functionality; human-perceived Coach Digest accuracy, relevance, and timing remain open for the planned pilot.

1.4 Scope

Table 1 separates completed work from claims that need more evidence.

Component	Capstone status	Claim limit
Profile construction	Experimental React application	The Best-Worst Scaling design is research-grounded but is not a validated Twinkl instrument.
Synthetic Journal Entries and LLM-Judge VIF Labels	Complete development corpus	The controls do not rule out generator or judge bias; correlated bias was not measured.
VIF Critic (Offline)	Completed capstone experiment programme	Results support an offline research contribution, not user-facing Drift authority.
Weekly Drift Detection	Development implementation with model and prompt comparisons	Accuracy was measured on AI-reviewed synthetic development data. Later prompt changes have not been assessed against a fresh final test.
Coach Digest	Experimental, with saved generation and AI editorial review	Automated validation and AI review support development; human-perceived quality remains untested. The five-response scored sample is separate from later generations.
North Star Moment	AI evaluation integrated in saved replay	The 501-week comparison and targeted September update favour full eligible history for Card precision and Opportunity recall. The saved replay uses completed full-history outcomes for five Personas and 27 reviewed weeks. These are AI assessments of synthetic histories. Live execution requires a valid separately configured budget, and the final source-level walkthrough remains pending.

Component	Capstone status	Claim limit
Experience and Inspect	Core assessment path implemented and tested; capstone application in progress	Automated tests and browser walkthroughs verify selected application behaviour. Coach Digest feedback capture, longitudinal Core Value history, the final professor walkthrough, real-user study, and optional live rerun remain open.

Table 1. Capstone implementation status and the evidence boundary for each component.

1.5 Academic contribution and practice-module alignment

Twinkl combines an R&D investigation with an integrated proof of concept. Its contribution is a staged study of value-grounded accountability: a research-based Profile, the entry-level VIF Critic (Offline), the longitudinal Weekly Drift Reviewer and deterministic Drift Detector, and a React application that exposes the evidence behind each result. Table 2 maps this contribution to the four practice modules identified in the April proposal, using the current architecture rather than the proposal’s historical runtime design.

Practice module	Question addressed in Twinkl	Current implementation and evidence
Pattern Recognition Systems	Can unstructured Journal Entries support ordinal value-alignment prediction under severe class imbalance?	The VIF Critic (Offline) study evaluates embeddings, ordinal and long-tail losses, uncertainty, persona-level splits, three training seeds, and Conflict recall. Investigation 2 reports the empirical frontier and negative results.

Practice module	Question addressed in Twinkl	Current implementation and evidence
Intelligent Sensing Systems	Can longitudinal text act as a behavioural signal relative to a confirmed Profile?	Chronological Journal Entry ordering, Monday-to-Sunday week eligibility, cumulative history recomputed at each weekly cutoff, and displayed-nudge classification from observable content treat text and time as a supporting sensing stream. A Journal Entry gap produces Insufficient Evidence only when it blocks a claim after recent Conflict evidence. The capstone remains text-only and does not claim a novel or multimodal sensing contribution.
Intelligent Reasoning Systems	Can semantic evidence and explicit rules produce a contestable Drift conclusion?	The Weekly Drift Reviewer decides Conflict, Not Conflict, or Abstain; the deterministic Drift Detector applies the two-consecutive-Conflict rule; Coach Digest receives structured Weekly Drift Detection output without deciding Drift.
Architecting AI Systems	Can model interpretation and deterministic rules be integrated into an inspectable application?	The React Experience, Python service, and Inspect connect prompts and recorded responses to validated application decisions. Automated tests check evidence preservation and failure recovery, while saved model receipts support reproduction of the reported studies.

Table 2. Current evidence mapped to the four Intelligent Systems practice modules.

2. Related Work

2.1 Human values and Profile construction

The original Theory of Basic Human Values defines ten broad values and their motivational relations [5]; later work refines these into 19 narrower values while preserving the circular motivational continuum [6].

Twinkl uses the original ten-value model as a stable vocabulary. It does not treat a Profile as a diagnosis or fixed identity.

Best-Worst Scaling asks a person to choose the most and least important items from subsets and estimates relative importance. In Lee *et al.*'s comparison, the Schwartz-values Best-Worst survey took significantly less respondent time than the traditional Schwartz Values Survey [7], while Marley and Louviere formalise probabilistic best-worst choice models [8]. Twinkl uses a balanced design in which each item appears equally often and each pair occurs together equally often; the resulting Profile is a declared reference for later reflection, not an observed-behaviour ground truth.

2.2 Reflection, accountability, and longitudinal personal data

Personal informatics research describes preparation, collection, integration, reflection, and action as connected stages [1]. Later work recasts self-tracking as an ongoing process that includes lapsing and potential resumption among people with varying goals [2]. This motivates Twinkl's longitudinal design; Insufficient Evidence is a separate decision rule for failed reviews or evidence gaps that block a current Drift claim.

Reflective informatics identifies breakdown, inquiry, and transformation as useful dimensions for reflection [3]. Recent contextual AI journaling work demonstrates a complementary design: MindScope combines passively sensed behaviour with LLM-generated prompts and reports an eight-week exploratory study with 20 college students [11]. This provides exploratory evidence that longitudinal context can be used to personalise reflection prompts, but MindScope's purpose is well-being-oriented prompting rather than comparison with a user-confirmed Profile.

LLM personalisation research addresses adjacent technical problems. Wu *et al.* infer unspoken preferences through multi-turn interaction so an assistant can tailor later responses [12], while Tan *et al.* use cited evidence to improve retrieval from long-term dialogue memory [13]. These studies improve how an assistant remembers or adapts to the user. Sycophancy research, however, shows why response fit alone is not sufficient: an assistant can mirror a user's beliefs when a more truthful response would disagree [4].

The resulting research gap is narrow. Twinkl does not claim novelty for journaling, Profile construction, long-term memory, or AI-generated reflection in isolation. It investigates their integration into a contestable accountability path: user-confirmed Core Values provide the reference; chronological Journal Entries provide behavioural evidence; Weekly Drift Detection combines Weekly Drift Reviewer Decisions with the Drift Detector and abstention; and Coach Digest returns cited, non-prescriptive reflection. This path implements breakdown detection in a limited form and supports inquiry through one open question, without claiming that the interaction causes transformation or behaviour change.

2.3 Synthetic longitudinal data and LLM supervision

Synthetic text can make a controlled learning problem feasible when equivalent longitudinal human data and objective labels are unavailable to the project, but corpus volume alone is not evidence of diversity, realism, or label validity. Generate-annotate-learn methods demonstrate the value of separating text generation from pseudo-labelling and show that the effectiveness of synthetic examples depends on their fidelity and diversity [14]. For Twinkl, longitudinal generation adds a second requirement: variation across personas must coexist with continuity within each persona. Demographic and narrative controls therefore establish designed coverage, not population representativeness.

LLM-as-judge methods add scalable supervision, but they can exhibit position, verbosity, and self-enhancement bias [15]. Because one model family participated in both generation and labelling, Twinkl treats correlated bias as a possibility rather than a demonstrated effect in this corpus. Twinkl therefore preserves label rationales, compares a shared subset with project-team human annotation, measures repeated-call stability separately, and identifies the source of each label or review. These controls support bounded development use; they do not convert synthetic text or AI review into human ground truth.

The NIST AI Risk Management Framework describes valid and reliable, accountable and transparent, and privacy-enhanced AI as interrelated trustworthiness characteristics [16]. Twinkl does not claim conformance; it addresses a narrow subset of related concerns through frozen holdouts, explicit acceptance gates, separation of generation metadata from production-like inputs, fail-closed model validation, and recorded model receipts. These controls reduce avoidable ambiguity without granting deployment approval.

2.4 Ordinal learning and long-tail labels

Each LLM-Judge VIF Label is ordinal: -1 for Conflict, 0 for neutral, and $+1$ for alignment. The distance between the classes matters. QWK is a chance-corrected agreement measure in which quadratic weights penalise larger ordinal disagreements more than adjacent disagreements [17]. Ordinal methods such as CORAL model ordered classes directly [18]. Balanced Softmax incorporates class frequency into the softmax objective to address long-tailed label distributions [19]. Monte Carlo Dropout gives an approximate uncertainty measure through repeated stochastic predictions [20].

These methods are relevant because neutral labels dominate most value dimensions. Accuracy alone can therefore look strong while Conflict recall remains poor.

3. Method

3.1 Study design and evidence boundaries

The study combines three core empirical investigations, subsequent system studies, and an implementation objective. Investigation 1 examines corpus construction, targeted train-only augmentation, and agreement on individual LLM-Judge VIF Labels. Investigation 2 uses persona-level train, validation, and test splits to assess the VIF Critic (Offline). Investigation 3 evaluates Weekly Drift Detection against frozen, AI-reviewed synthetic development references. Later studies examine sensitivity to Core Value definitions and North Star Moment quotation selection. Application verification tests deterministic rules and failure recovery, while Coach Digest AI review examines source interpretation and wording. Human agreement on individual Journal Entries does not validate longitudinal Drift, and neither AI-reviewed development references nor working replays measure real-user usefulness.

Investigation and report use	Data or experiment setup	Sample size	Known Drifts	Model	Reasoning effort	Repeats
Investigation 1: label agreement	Shared human-overlap benchmark	115 Journal Entries; 19 personas	—	Original persisted LLM-Judge model not retained; three project-team annotators	Not retained	One blind-first annotation pass per annotator
Investigation 1: targeted data lifts	Frozen-holdout train-only augmentation	24 personas; 191 Journal Entries added across two batches	—	Balanced Softmax VIF Critic (Offline); Nomic embedding [21]	—	Three training seeds per model family
Investigation 2: VIF Critic (Offline) reference	Corrected persona-level split	1,022 train; 217 validation; 221 test Journal Entries	—	Balanced Softmax VIF Critic (Offline); Nomic embedding	—	Three training seeds
Investigation 3: VIF hand-off ablation	Frozen hand-off development union	106 cases; 894 Journal Entry/Core Value combinations	33	gpt-5.4-mini-2026-03-17	None	Three
Investigation 3: adopted Weekly Drift Reviewer model	Complete development data	292 cases; 951 Persona-week prompts per repeat	42	gpt-5.6-luna	Low	Three
Core Value definition study	Matched prompts with and without definitions	204 Personas; 951 weeks per repeat; 5,706 requests	Not accuracy-scored	gpt-5.6-luna	Low	Three per prompt setup
North Star Moment comparison	Nomic top three versus full eligible history	105 Personas; 501 weeks; 38 contexts recomputed in targeted update	—	Luna-low selection; Luna-xhigh AI evaluation	Low / xhigh	Paired methods; separate 20-case consistency sample
Historical Coach Digest Evals	August Persona key weeks	Five accepted responses with same-model AI scores	—	gpt-5.6-luna	None	One accepted response per Persona

Investigation and report use	Data or experiment setup	Sample size	Known Drifts	Model	Reasoning effort	Repeats
Current Coach Digest development evidence	September base responses and source-context comparison	27 base responses; 22 pairs containing 44 responses	—	gpt-5.6-luna	None	Accepted outputs include recorded repairs; no new Coach Digest Evals

Table 3. Evidence datasets and experiment setups. Cases count Persona/Core Value histories; Persona-week prompts count reviews at weekly cutoffs. Known Drifts apply only to the accuracy studies.

The sequencing in Table 3 is methodologically important. The synthetic corpus was constructed and labelled before it became training evidence. Later targeted batches were added only to the training partition after the validation and test personas had been frozen. The VIF hand-off ablation then preceded the complete-development Luna study and tested whether VIF Critic Predictions improved a gpt-5.4-mini-2026-03-17 Weekly Drift Reviewer. The later evidence established gpt-5.6-luna at low reasoning effort, without VIF Critic input, as the fixed Weekly Drift Reviewer contract. No matched Luna-low VIF ablation was run.

3.2 Architecture

Figure 1 shows the adopted architecture. Onboarding creates the Profile, after which the application records Journal Entries and optional displayed-nudge responses. For each Core Value, the Weekly Drift Reviewer examines cumulative Journal Entry history, and the Drift Detector applies those decisions plus review and evidence-gap status to produce Active Drift, No Active Drift, or Insufficient Evidence. Coach Digest cites the saved evidence and asks one non-prescriptive question. Inspect connects these stages through a frontend prompt and decision trace: an assessor can examine the supplied instructions and evidence, the recorded response, its validation, and the deterministic processing behind the displayed result.

The April 2026 proposal positioned the VIF Critic (Offline) as the runtime model that would route weekly reflection. Subsequent evidence refined that design. The VIF hand-off ablation found no Drift-recall gain from adding VIF Critic Predictions to the tested gpt-5.4-mini-2026-03-17 input or scheduling setups, after which the complete-development comparison established direct Luna-low review as the adopted Weekly Drift Reviewer contract. The current architecture therefore gives user-facing Drift authority to the Weekly Drift Reviewer followed by the deterministic Drift Detector and retains the VIF Critic (Offline) as an independently evaluated research contribution. This is a deliberate evidence-driven allocation of component authority rather than an abandonment of the VIF Critic (Offline) investigation.

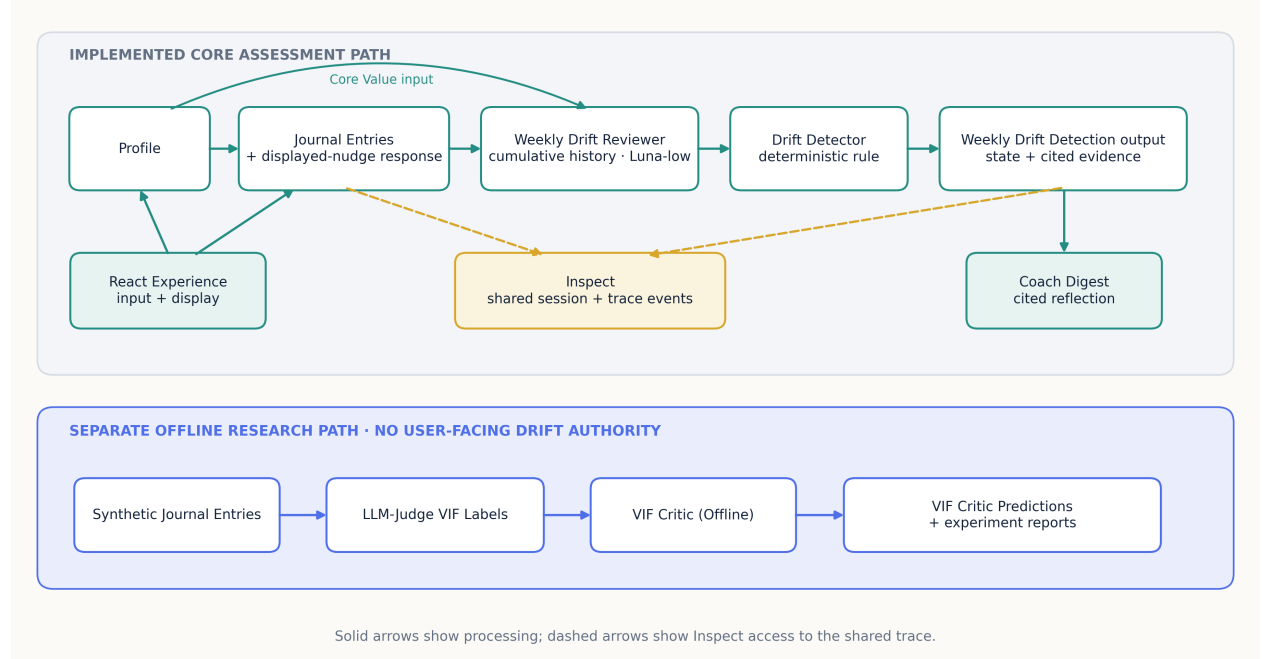


Figure 1: Adopted Twinkl architecture. The upper lane shows the implemented core assessment path and shared trace; the blue lane shows the separate offline research path. No arrow links VIF Critic Predictions to Weekly Drift Detection.

3.3 Profile construction

The Profile uses 11 value objects (assessment items) because Universalism has separate nature and social facets before the final ten-value merge. Across 11 sets of six objects, every object appears six times and every pair appears together three times; the user selects one Most and one Least item in each set.

For item i , the raw Best-Worst Scaling score is

$$s_i = \frac{B_i - W_i}{6}, \quad -1 \leq s_i \leq 1,$$

where B_i and W_i are the Most and Least counts. Twinkl takes the mean of the two Universalism facet scores. It then subtracts the lowest of the ten scores from every score, adds one, and normalises the ten shifted scores to sum to one. This transformation preserves score ordering; a weight twice as large does not mean that the value is twice as important. The value or values with the highest score are shown for confirmation. If more than two values tie at the top, the user must select exactly two. A confirmed Profile therefore has at most two Core Values.

The onboarding Experience presents one balanced group at a time without exposing Schwartz labels or scores. Figure 2 shows the six selection cards and the separate Most and Least choices for one group. After confirmation, Inspect exposes the complete choice-to-Profile calculation: an assessor can follow the 22 recorded selections through object counts, the Universalism merge, the ten normalised weights, the confirmed Core Values, and the Python validation result. This deterministic path is complete before any model-assisted interpretation of Journal Entries occurs.

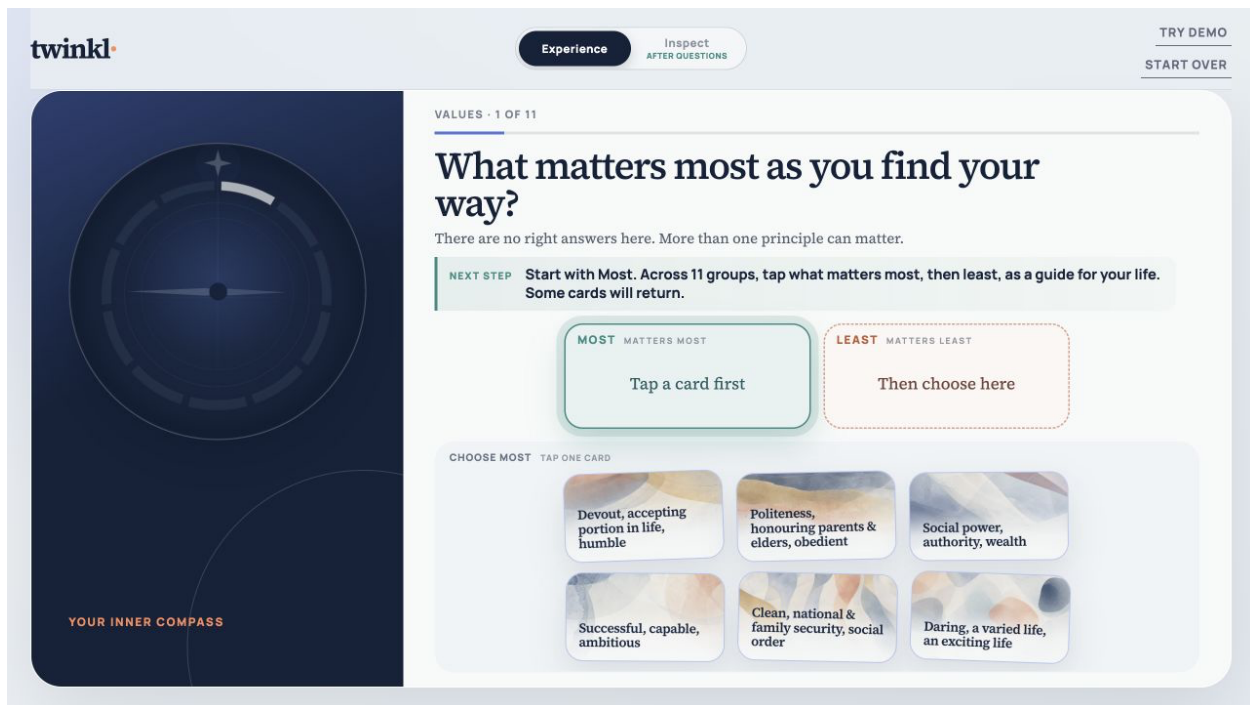


Figure 2: One onboarding group presents six selection cards and requires separate Most and Least choices without exposing Schwartz labels or scores.

The April proposal mock-up showed six forced-choice sets and a three-value summary. The implemented Schwartz Values Best-Worst Survey (SVBWS) design instead uses 11 balanced groups and confirms at most two Core Values. This change is a design refinement, not evidence that the Twinkl instrument is psychometrically validated.

3.4 Synthetic corpus and LLM-Judge VIF Labels

The corpus contains 204 personas and 1,651 Journal Entries. Designed diversity spans age range, profession, culture, tone, verbosity, and reflection mode. These configuration dimensions broaden controlled coverage but do not establish that the personas represent a real population. Generation ran in parallel between personas but sequentially within each persona; every new Journal Entry received the earlier entries for that persona so that events and relationships could persist over time.

Prompt design favoured emergent behaviour over prescribed value statements. Persona biographies expressed priorities through concrete life details, while Journal Entry instructions varied the type of moment without naming a Schwartz value. Banned terms reduced direct label leakage, and production-like decision logic received Journal Entry content and chronology rather than tone, reflection mode, or reference labels. Removing earlier dependencies on generation metadata was an important methodological correction: otherwise downstream logic could exploit information unavailable for a real Journal Entry.

Claude Code subagents created the original personas, Journal Entries, and LLM-Judge VIF Labels. The committed persona, Journal Entry, and label prompt templates are version 1.0.0. The original run files do not retain a stable Claude model identifier or model snapshot, which limits exact reproduction of those labels.

Later LLM-Judge studies have complete model and prompt receipts, but they do not replace the original persisted labels used in the corrected-split VIF Critic (Offline) reference.

For each Journal Entry, the LLM-Judge assigned ten ternary LLM-Judge VIF Labels and a short rationale for each non-zero label. Recomputing the persisted parquet file gives 16,510 labels: 12,535 neutral labels, 2,810 alignment labels, and 1,165 Conflict labels. Table 4 shows the resulting distribution. The 75.92% neutral share makes the long-tail problem visible before model training.

LLM-Judge VIF Label	Count	Share
Conflict (-1)	1,165	7.06%
Neutral (0)	12,535	75.92%
Alignment (+1)	2,810	17.02%
Total	16,510	100.00%

Table 4. Distribution recomputed from the persisted LLM-Judge VIF Labels in `judge_labels.parquet` (see Appendix A for the source path).

When model diagnostics exposed weak value dimensions, augmentation followed a five-step loop: **freeze, generate, verify, judge, and retrain**. The first train-only batch added 12 Power/Security personas and 95 Journal Entries. The second added 12 Hedonism/Security personas and 96 Journal Entries. Before either batch, the repository persisted the existing persona registry and validation/test holdout. Raw-batch verification then checked the new-persona count, entry-count bounds, intended value and tension coverage, registry state, and entries in both Unsettled (moments of giving ground) and other generation reflection modes. Acceptance depended on the resulting LLM-Judge VIF Labels rather than prompt intent alone. This design made each retrain attributable to new training evidence instead of a changed holdout.

The final parquet file contains 1,651 labelled Journal Entries, of which 1,594 rows contain rationale JSON. Des, JL, and KM, all members of the project team, independently labelled the same 115 Journal Entries from 19 personas. The annotation tool withheld the LLM-Judge comparison until each first-pass annotation was saved, reducing anchoring to the LLM-Judge result; this blind-first process is not independent external validation. Fleiss’ kappa measures agreement among the three humans [22], whereas Cohen’s kappa measures agreement between one human and the LLM-Judge [23]. We report the mean of the three LLM-Judge-human Cohen values and do not treat either statistic as a ceiling on the other.

3.5 VIF Critic (Offline)

The VIF Critic (Offline) began with a deliberately compact multilayer perceptron (MLP) rather than a larger sequence model. Its 23,454 parameters receive a 256-dimensional frozen `nomic-ai/nomic-embed-text-v1.5` embedding [21] and the ten normalised Profile weights. The historical corrected-split reference uses one Journal Entry at a time, two 64-unit hidden layers, dropout of 0.3, and 30 output logits. The logits represent three ordered classes for each of ten value dimensions.

The completed experiment archive spans 69 run IDs and 133 persisted MLP configurations. It covers ordinal, distance-weighted, class-balanced, margin-based, two-stage, and soft-label loss families; three frozen embedding families; recall-aware checkpoint selection; target repair; compact-history inputs; and legacy

weighted-MSE baselines. A Bayesian implementation and notebooks exist, but no Bayesian configuration appears in the persisted run archive. The persisted configurations are analysed as intervention families rather than 133 independent discoveries because many share data, architecture, or selection decisions.

The split is by persona. The historical corrected-split reference predates the two targeted synthetic batches and therefore contains 1,460 Journal Entries: 1,022 training Journal Entries, 217 validation Journal Entries, and 221 test Journal Entries. The later batches expanded only the training partition to 1,213 Journal Entries while retaining the frozen validation and test partitions; the corresponding retrains did not replace the historical reference. The split seed is 2025. The three training seeds are 11, 22, and 33. The Balanced Softmax reference uses a learning rate of 0.015522, weight decay of 0.01, batch size 16, at most 100 epochs, and early stopping patience 20. Fifty Monte Carlo Dropout samples provide the uncertainty diagnostic.

QWK is the main ordinal-agreement measure. For each value, Conflict recall is the proportion of reference -1 labels that the VIF Critic (Offline) predicts as -1. Aggregate QWK and Conflict recall average the per-value scores, giving equal weight to each value with a defined score. The error-uncertainty correlation is the Spearman rank correlation between absolute prediction error and Monte Carlo Dropout uncertainty; a larger positive value means that larger errors tended to receive higher uncertainty. It is an uncertainty diagnostic rather than a complete calibration measure. We also inspect class-specific recall and the share of Monte Carlo Dropout mean predictions strictly between -0.3 and +0.3 because a high QWK can hide weak Conflict detection.

3.6 Weekly Drift Detection

For each Persona-week, the Weekly Drift Reviewer receives the Persona's cumulative Journal Entry history through the weekly cutoff and all Core Values under review, rather than hidden generation or labelling metadata. It returns Conflict, Not Conflict, or Abstain (no classification) for each current-week Journal Entry/Core Value combination. The fixed development model contract uses `gpt-5.6-luna` with the provider's reasoning-effort setting at low, structured output, a 2,000-output-token limit, `store: false`, and fail-closed validation.

The Drift Detector owns the sequence rule. It counts each uninterrupted run of at least two consecutive Conflicts for the same Core Value as one Drift, regardless of its length. In the historical consensus-label analysis, any transition into -1 occurred in 102 of 292 Core Value trajectories, representing 92 of 204 personas. Two consecutive Conflicts occurred in 41 of 292 trajectories and represented 40 personas; three consecutive Conflicts occurred in 20 trajectories and represented 20 personas; and four consecutive Conflicts occurred in five trajectories. No persona count was recorded for the four-Conflict result. These descriptive results informed the two-Conflict design; the threshold remains a project-defined rule rather than a validated behavioural cutoff.

Figure 3 makes the implemented state rule explicit. A first valid Conflict starts a run while the current state remains No Active Drift. A second adjacent valid Conflict creates Active Drift. A failed review always produces Insufficient Evidence, whereas a valid Abstain or Journal Entry gap does so only when it blocks a current claim after recent Conflict evidence or an unresolved state. Historical Drift Records remain stored after the current state changes.

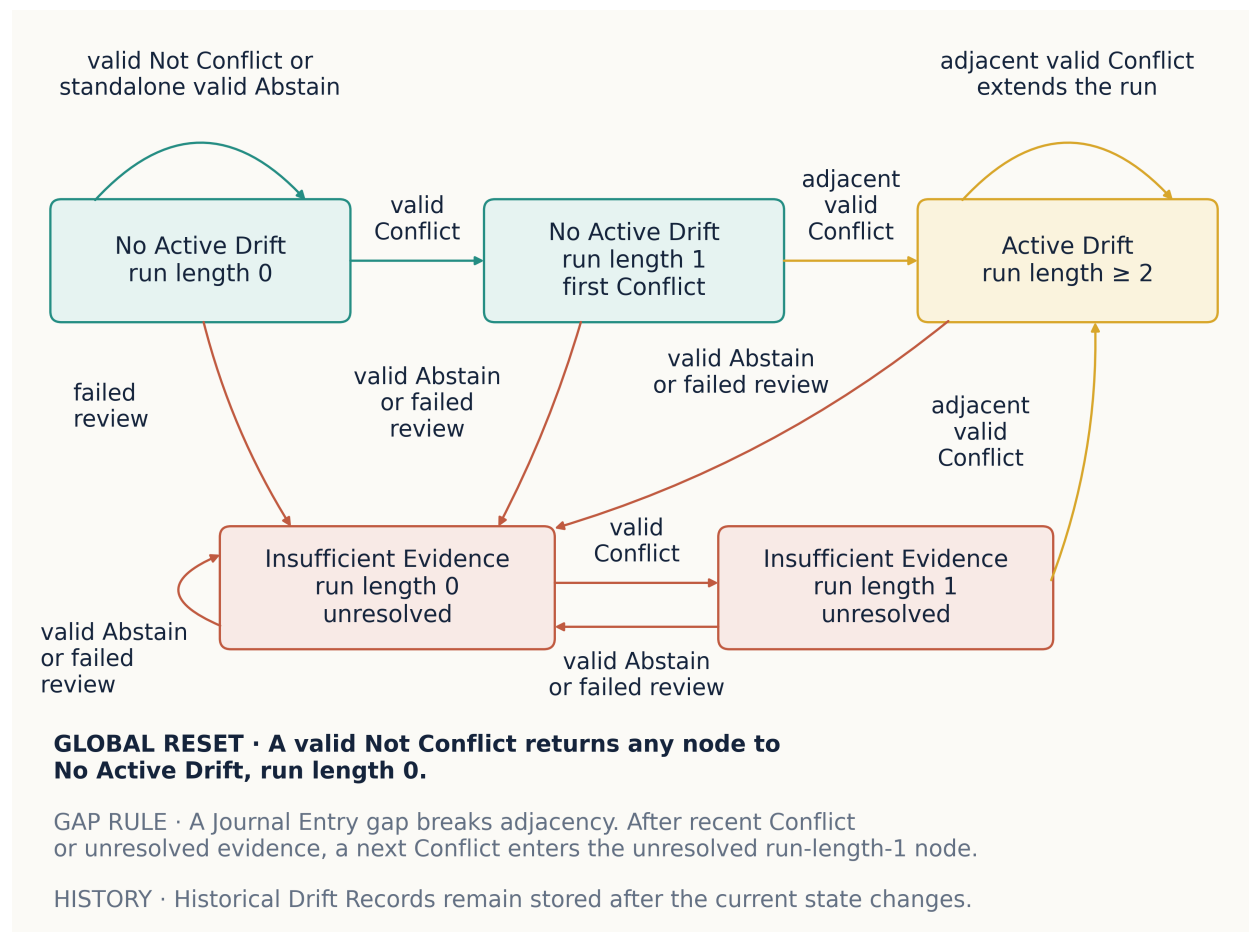


Figure 3: Drift Detector transitions for one Core Value. The five nodes distinguish the current state and the run-length or unresolved substate that determines the next transition.

The complete development review contains 292 resolved cases. A resolved case is one Persona/Core Value history with a final Drift reference outcome and no open review decision. These cases contain 2,377 Journal Entry/Core Value combinations and 42 Drifts across 36 histories with at least one Drift. Two isolated AI reviews using `gpt-5.6-sol` at xhigh reasoning effort covered the 186 cases outside the earlier frozen 106-case set. They agreed on 95.2% of 1,483 decisions, although they initially identified 32 and 85 Conflicts respectively. The high overall agreement therefore coexisted with different rates of identifying the rare Conflict class. A disagreement-only review resolved the remaining 71 decisions. The earlier frozen set used the same review approach; four prior Uncertain decisions were later reviewed with `claude-opus-4-8`. These are AI-reviewed LLM-Judge Conflict Labels, not human validation.

Drift recall is the proportion of reference Drifts matched by a predicted Drift; Drift precision is the proportion of predicted Drifts matched to a reference. Matching is one-to-one within the same Persona and Core Value: a predicted confirmation must fall between the reference onset and two Journal Entry positions after the reference run ends. Unmatched predicted Drifts count as false Drift alerts, including additional unmatched predictions in a history that contains reference Drift. Detection delay measures days from reference confirmation to the matched alert.

Each complete-development Weekly Drift Reviewer setup used 951 Persona-week prompts and three repeats. Coverage is measured over Persona/Core Value histories: a history is covered when the returned decisions establish an adjacent pair of Conflicts, or when every observed adjacent Journal Entry pair contains a definite Not Conflict decision that rules out such a pair. This differs from entry-level coverage, the proportion of individual decisions returned as Conflict or Not Conflict. Paired intervals use 10,000 trajectory-level bootstrap resamples. The VIF hand-off study uses base seed 752,520,000, while the complete-development operating-point study uses base seed 5,256,000; each comparison adds a fixed offset to its base seed. Resampling keeps decisions from the same Persona/Core Value history together. Paired differences are the median of the three within-repeat differences and therefore need not equal the difference between the separately reported medians.

3.7 VIF Critic hand-off ablation

Before the Luna-low contract was adopted, the hand-off study tested three gpt-5.4-mini-2026-03-17 setups on a frozen development union with 33 known Drifts across 106 cases:

- Weekly Drift Reviewer without VIF Critic input;
- Weekly Drift Reviewer with raw VIF Critic Predictions;
- VIF-Critic-triggered early Weekly Drift Reviewer calls plus Weekly Drift Detection.

All setups used no reasoning effort and three repeats. Only the VIF Critic input or schedule changed. The early trigger required two consecutive Journal Entries with mean predicted Conflict probability $P(-1) \geq 0.8$ and maximum uncertainty no greater than 1.010153. This experiment tested whether the VIF Critic (Offline) improved downstream Weekly Drift Detection with gpt-5.4-mini at no reasoning effort. It did not test whether the VIF Critic (Offline) alone could replace Weekly Drift Detection, and it did not test VIF Critic input under the later Luna-low contract.

3.8 Coach Digest validation and development evaluation

Coach Digest receives structured Weekly Drift Detection output, including prior-week evidence when a state comparison is available. It must cite relevant Journal Entry text, avoid prescriptive instructions and unsupported claims, and ask one open question. When further reflection is needed, it states the evidence limit without deciding whether Drift exists. Coach Digest Validations check every extracted quotation against the supplied evidence and screen for unsupported state claims, technical jargon, excessive length, and specified language problems. These checks can verify quotation matches and question form, but assessing whether a question is open, relevant, and non-prescriptive requires semantic review.

Coach Digest Evals use four five-point scores: correctness, evidence specificity, non-prescriptive tone, and tension honesty (whether the response accurately represents the supplied Drift state and evidence limits). The target mean is at least 3.5 for each score. The evaluator also checks whether the question is open and relevant. The same gpt-5.6-luna model at no reasoning effort generated and evaluated the five August responses. This creates a risk of self-enhancement bias in the AI review [15]; the study did not measure whether that bias affected these five responses.

Later development produced 27 base responses for the current Persona roster and 22 pairs generated with and without selected North Star Moment source context. Each pair began with the same instructions and weekly

evidence, apart from the additional source context. Some responses subsequently required different repair instructions, so the pairs support qualitative comparison rather than a controlled estimate of benefit. The saved replay displays the 22 responses without additional context and five base responses for weeks without a selection. These later generations received AI editorial review but no new Coach Digest Evals scores.

A separate behavioural regression corpus contains eight pairs of Codex-authored synthetic examples covering quotation grounding, prescriptive language, unsupported interpretation, and instruction-like Journal Entries. Expected judgements remain outside model inputs. Automated tests check input preparation and structural failures; semantic evaluation of these examples remains future work.

3.9 North Star Moment: implementation and paired AI evaluation

North Star Moment (NSM) is implemented to quote a reported action or already-made choice supporting a confirmed Core Value after an eligible closed-week result. Active Drift uses writing from before onset; No Active Drift prefers writing from the reviewed week and otherwise uses an earlier reference point. The implementation supports source-checked cards and Inspect records in both frontend paths. Saved replay uses the completed full-history outcomes for five selected Personas and 27 reviewed weeks from the targeted September update, while live execution remains subject to its separately configured budget.

Figure 4 shows the recorded assessment of one selected source. Inspect exposes the reported action, its relation to the Core Value, and whether the source also contains opposing behaviour, allowing the quotation to be examined alongside the basis for its selection.

The [NSM experiment methodology](#) compares Nomic top-three retrieval followed by Luna-low review with Luna-low review of the full eligible history. Each retrieval candidate contains one complete Journal Entry and its eligible Persona response. The paired comparison covers 501 closed weeks from 105 synthetic Personas, with 81 development and 24 qualified-final Personas. A shared Luna-xhigh evaluator independently assesses complete eligible histories and the exact quotations selected by each method. Card precision measures correct cards among scored displayed cards; Opportunity recall measures weeks receiving a correct card among scored reference opportunities. The fixed contradiction-recheck rule and common exclusions preserve comparability when AI judgments are unresolved, while 10,000 paired whole-Persona bootstrap resamples account for correlated weeks within each partition.

Following changes to Weekly Drift Detection, a [targeted update](#) identified 38 changed NSM contexts across 22 Personas, comprising 29 development weeks and nine final weeks. Both NSM methods were recomputed for those contexts, while 463 unchanged contexts retained their observations, including failures and unresolved judgments. Reuse required matching inputs, model settings, and evaluation procedures. The original 20-case consistency sample was unaffected and retained. All 501 cases were then regraded; these results constitute a targeted update with retained observations rather than a fresh independent experiment.

3.10 Software verification and research reproducibility

Software verification follows the responsibilities of the application components. Unit tests exercise Profile scoring and Drift state transitions, while integration tests check evidence preservation and session consistency. Simulated provider failures test response handling and retry behaviour, and browser tests exercise the React application and Python service with controlled model responses. These methods assess application behaviour independently of semantic model quality. Table 5 summarises the principal checks.

3. Follow the AI assessment

The recorded model response answers three questions for every source. These are the runtime AI assessments used to derive the Moment, not human validation or a separate benchmark evaluator's verdict.

Making the world a fairer, better place

Last recorded attempt: Completed.

5fa8b540:entry:4

Selected quotation

What did the writer do?

The writer stayed back and helped the peon uncle repair a broken classroom desk.

How does the action relate to this Core Value?

This is direct practical assistance that supports another person's welfare.

Does this source also show opposing behavior?

No supplied writing reports behavior by the writer against the welfare of people or nature.

AI reason

Observable Choice

Stayed back to help the peon uncle fix a broken desk in my classroom.

Figure 4: Inspect's recorded source assessment for Nisha's fourth-week North Star Moment. The selected quotation describes an action before Drift onset. This saved synthetic replay illustrates the inspection process; the assessment is AI-generated.

Concern	Required behaviour	Verification evidence
Profile and Drift rules	Deterministic scoring, chronological cutoffs, cross-week Drift, and explicit evidence-gap states	Profile, Drift Detector, and service tests
Research data integrity	Unique identity keys and joins that cannot conceal missing Journal Entries	Duplicate-key and merge-integrity regressions
Evidence and persistence	Every quoted passage is grounded; reloading preserves prior-week comparisons and validation context	Mixed genuine/fabricated quotation cases and Parquet round-trip tests
Model calls	Retries follow the recorded policy; refusals and incomplete responses are withheld	Simulated failures through the provider client library
Experience and Inspect	Selected-week context survives reload; retry does not repeat completed review; deletion clears matching state	React integration tests and two Chromium journeys against the Python API
Historical reproduction	Reported results remain verifiable against their original implementation	Recorded source revisions and model receipts

Table 5. Application and research verification methods.

Historical experiments are checked against their recorded source revisions so that later application changes do not alter the basis of reported results. Current regression tests assess the maintained implementation. Saved requests and model receipts support inspection and re-scoring, although a fresh model call may produce a different response.

The public Railway assessment at <https://onboarding-production-1dd2.up.railway.app/> provides an anonymous, assessment-only application. Saved replays require no provider key, while live manual use can trigger paid provider calls. Local verification and saved evidence support the System Implementation & Demo assessment; they do not establish the state of every hosted revision or provide service-level guarantees.

3.11 Inspect as prompt and decision telemetry

Inspect allows an assessor to examine how a displayed result was produced. Its prompt telemetry records the supplied instructions and Journal Entry context alongside model settings, recorded responses, and validation outcomes. Organised around the selected week, these records connect model interpretation to the deterministic processing that follows. An assessor can therefore trace a Drift conclusion or Coach Digest from the available evidence through to the application decision.

The application records the messages assembled for the model, keeping application instructions separate from user-supplied content. This allows reviewers to check whether relevant evidence was included and whether a later repair changed the request. The trace describes observable processing and does not expose the model's internal reasoning.

North Star Moment inspection links eligible writing to source assessment and quotation selection, while the saved Coach Digest comparison shows how the supplied context and any repair instructions relate to each response. Where recorded, generation receipts also report token use, latency, and calculated cost. These saved records can be examined without making further model calls.

Figure 5 shows how Inspect pairs the recorded prompt with its associated response. The visible instructions establish the intended reflective tone, while the response can be checked for its use of the supplied evidence and question form.

Without North Star Moment: exact prompt and associated response

WITHOUT NORTH STAR MOMENT: EXACT ACCEPTED PROMPT

Message contract: live-prompt-boundary-v1

TRUSTED INSTRUCTIONS

You produce Twinkl's Coach Digest response.

Write a short, warm reflection to the person whose writing you have read, like a thoughtful colleague who remembers what matters to them. Address them as you.

You are not a therapist or a clinical assessor.

Stay reflective, not prescriptive.

Do not use scoring, alignment, or gamification jargon, or judgmental language. Specifically, never write the words "score", "scores", "scored", "alignment", "aligned", "misaligned", or the token "mean=" anywhere in the reflection.

Avoid giving advice, action plans, or micro-habits.

Associated Coach Digest response

After a long day, you kept quiet with Sharma sir because he was senior and you had no energy to start something. Then, feeling tired at the Hazratganj shop, you chose speed and spent twice as much at a chain store, even though you could have waited; you wrote, "I just wanted to get it done fast."

These two choices point to a repeated pull away from making the world a fairer, better place: staying silent when something may have needed saying, and choosing convenience over a more considered purchase. At the same time, you stayed back to help the peon uncle fix the broken desk, so the picture includes both care for people around you and moments when tiredness narrowed your choices.

When you think about staying quiet with Sharma sir and shopping quickly in Hazratganj, which moment feels more important to understand?

Figure 5: Inspect pairs the recorded Coach Digest prompt with its associated response for Nisha's fourth week. The capture shows the opening of the scrollable instructions and the complete response generated without additional North Star Moment context. Saved synthetic replay.

The detail available varies by component: some records contain parsed responses, and execution measurements are not complete for every event. Inspect therefore supports evidence review and diagnosis without constituting a complete provider log or a controlled performance benchmark.

4. Results

4.1 Investigation 1: corpus controls support bounded model development

The corrected-split corpus began with 180 personas and 1,460 Journal Entries. The two frozen-holdout targeted batches increased only the training partition, producing the final 204-persona, 1,651-Journal-Entry corpus shown in Figure 6. The first batch added 12 Power/Security personas and 95 Journal Entries. Its label quality-assurance (QA) sample contained seven retained examples, one ambiguous example, and no bad label. On the unchanged test set, median Power Conflict recall rose from 0.125 to 0.313 and Security Conflict recall remained 0.571. Aggregate median Conflict recall rose from 0.313 to 0.342, but QWK fell from 0.362 to 0.349 and Hedonism regressed. The batch demonstrated a local Power benefit without displacing the general reference.

The second batch added 12 Hedonism/Security personas and 96 Journal Entries under value-family and polarity acceptance gates. In the resulting Balanced Softmax family, Hedonism QWK improved modestly from 0.247 to 0.256, while Security QWK fell from 0.297 to 0.199. Aggregate median QWK was 0.346 and Conflict recall was 0.328, compared with 0.362 and 0.313 for the historical reference. The targeted data therefore changed particular decision boundaries but did not create a clean family-wide improvement. More synthetic data was not automatically better; the frozen comparison exposed both transfer and regression.

Human-human Fleiss' kappa was 0.56 on the shared benchmark, while mean LLM-Judge-human Cohen's kappa was 0.66. These measures answer complementary validation questions. Fleiss' kappa asks how consistently the three project-team annotators applied the task, whereas mean Cohen's kappa asks how closely the saved LLM-Judge VIF Labels overlapped with each annotator's interpretation. Because the rater structures differ, the numerical gap is descriptive rather than a paired advantage, and neither statistic defines a human-consistency ceiling.

Figure 7 shows the per-dimension coefficients separately. The mean LLM-Judge-human Cohen value was numerically larger than the human-human Fleiss value in nine of ten dimensions. Power was the exception, with 0.60 against 0.61; Universalism had the largest coefficients, while Conformity (0.43), Self-Direction (0.44), Achievement (0.47), and Security (0.48) had the weakest human-human agreement. The figure is therefore useful for locating dimensions that require more careful supervision, not for ranking humans against the LLM-Judge.

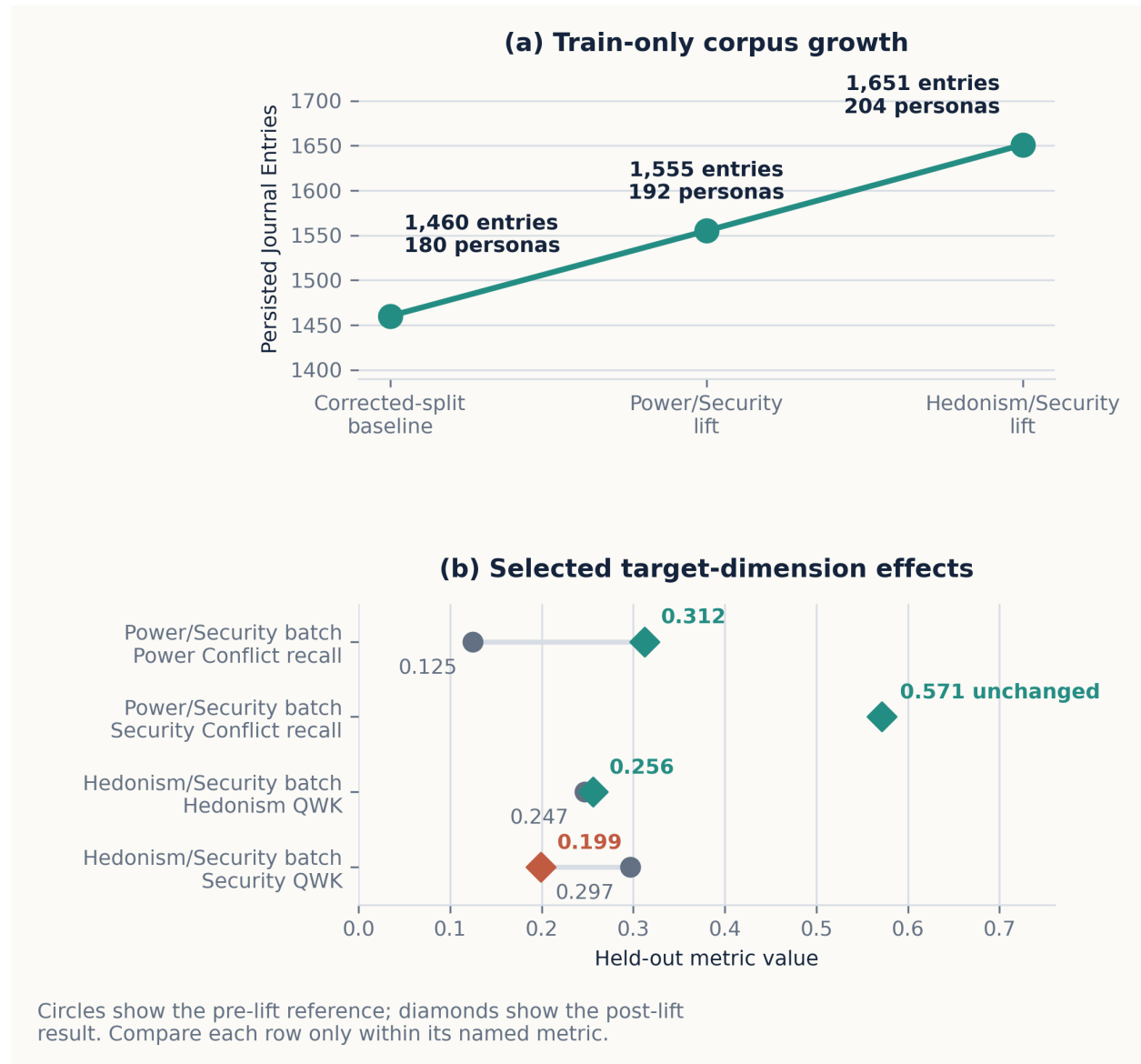


Figure 6: Growth and outcomes of the two train-only targeted synthetic batches. Panel (a) shows the frozen-holdout corpus progression. Panel (b) shows selected within-metric target-dimension changes; QWK and Conflict recall rows are not compared with each other.

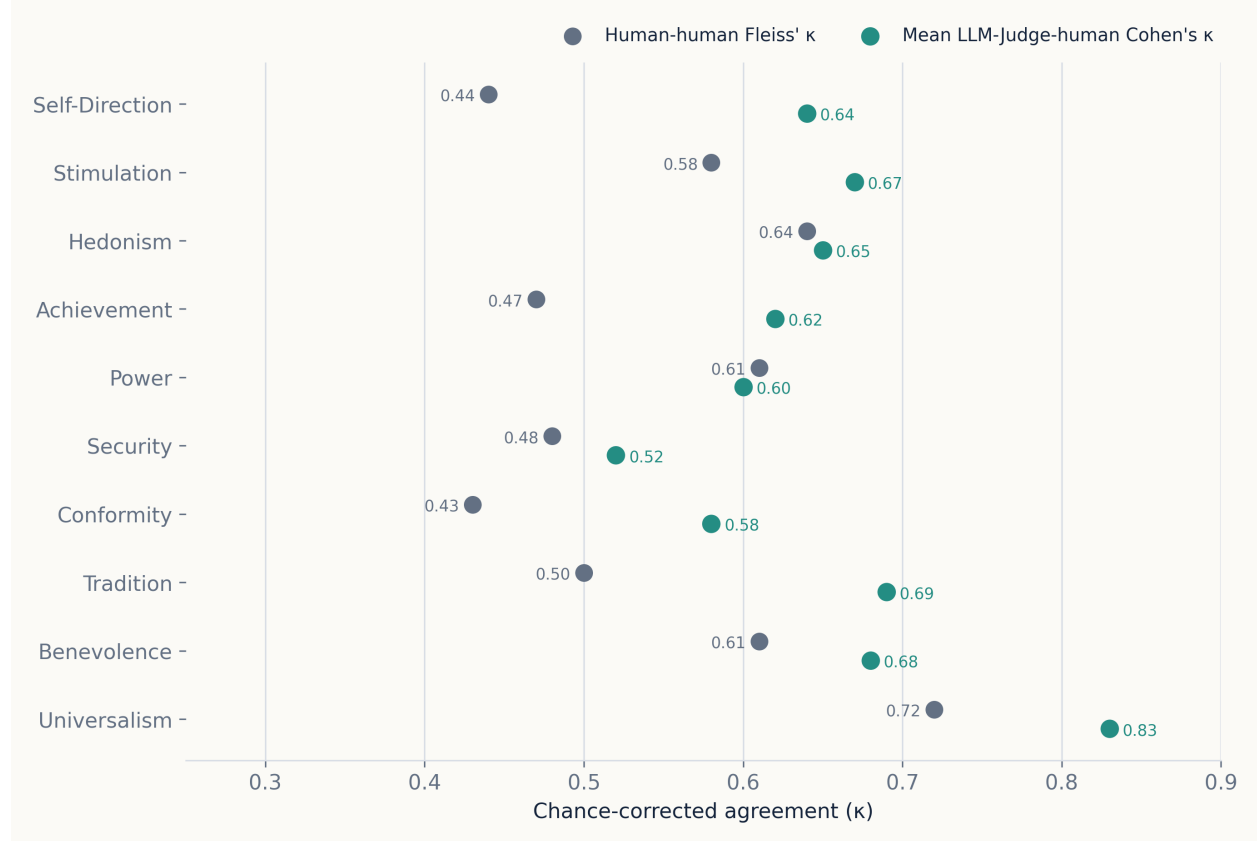


Figure 7: Chance-corrected agreement on the shared 115-Journal-Entry benchmark. Grey markers show human-human Fleiss' kappa; green markers show mean LLM-Judge-human Cohen's kappa. Un-connected markers preserve the separate rater structures while showing which value dimensions have weaker overlap.

A separate five-pass LLM-Judge study found per-dimension repeated-call Fleiss' kappa from 0.775 to 0.890. Its consensus labels changed the frozen holdout and did not become the active VIF Critic (Offline) target, so this stability result complements rather than extends the human-overlap benchmark. Taken together, the evidence supports bounded development use of the persisted labels without treating every dimension as equally reliable or the labels as human ground truth.

Investigation 1 finding: Explicit generation controls, frozen train-only augmentation, label QA, repeated-call stability, and the human-overlap benchmark make the synthetic corpus suitable for bounded model development. They do not establish population realism, objective labels, or independent human validation. The mixed targeted-batch results show why corpus quality must be assessed by held-out effects rather than volume or prompt intent alone.

4.2 Investigation 2: the compact VIF Critic (Offline) reaches a measurable but inadequate frontier

Table 6 reports the three corrected-split Balanced Softmax seeds. The family median, defined as the median across training seeds 11, 22, and 33, was 0.362 QWK and 0.313 Conflict recall. Seed 22 had the highest QWK and Conflict recall, but the spread across seeds shows why reporting only the best run would overstate stability.

Training seed	Test QWK	Conflict recall	Error-uncertainty correlation	Near-neutral MC-mean rate
11	0.362	0.277	0.727	0.642
22	0.378	0.342	0.713	0.621
33	0.358	0.313	0.655	0.565
Median	0.362	0.313	0.713	0.621

Table 6. Corrected-split Balanced Softmax VIF Critic (Offline) results across three training seeds. The near-neutral MC-mean rate is the share of Monte Carlo Dropout mean predictions strictly between -0.3 and $+0.3$; it is not the deterministic neutral-class prediction rate. Error-uncertainty correlation is an uncertainty diagnostic, not a complete calibration measure.

Balanced Softmax moved the VIF Critic (Offline) away from an all-neutral failure mode and recovered some rare Conflict labels, which is technically useful in a corpus where 75.92% of labels are neutral. However, the family-median result still misses about two thirds of reference Conflicts on average across value dimensions and therefore cannot support user-facing Drift authority.

The broader programme tested whether that limit came from the loss, representation, labels, selection rule, data support, or missing history. Ordinal-loss and long-tail-loss families reduced particular forms of neutral collapse, the tendency to predict neutral regardless of the reference label. The controlled Qwen encoder rerun reached slightly higher median QWK of 0.370 but remained weak on Hedonism and Power. Two-stage reformulation reduced Conflict recall, while consensus-label retrains changed the evaluation target. Selecting checkpoints by Conflict recall did not improve the persisted-label frontier, and adding a short Journal Entry history failed the criteria for a larger follow-up study. Across 69 run IDs and 133 persisted configurations, no intervention produced a stable all-metric replacement for the corrected-split Balanced Softmax reference under the original label definition.

Panel (a) of Figure 8 shows that the plateau was not uniform across Schwartz values. Median Conflict recall ranged from 0.125 for Power to 0.733 for Self-Direction. Achievement reached 0.286 and Stimulation 0.333, whereas Hedonism reached 0.652 and Security 0.571 on this historical target. Support varied from only three Universalism Conflicts to 45 Self-Direction Conflicts, so an apparently high or low recall can be unstable. Recall also does not report false-positive behaviour and must be read with QWK, precision, and seed spread. Later target-repair and hard-set studies still found unresolved Hedonism and Security boundaries, demonstrating that difficulty depends on the label regime as well as the value name. Panel (b) reports the later Luna-low study and is interpreted in Investigation 3; the aligned panels support descriptive error analysis, not a direct model comparison.

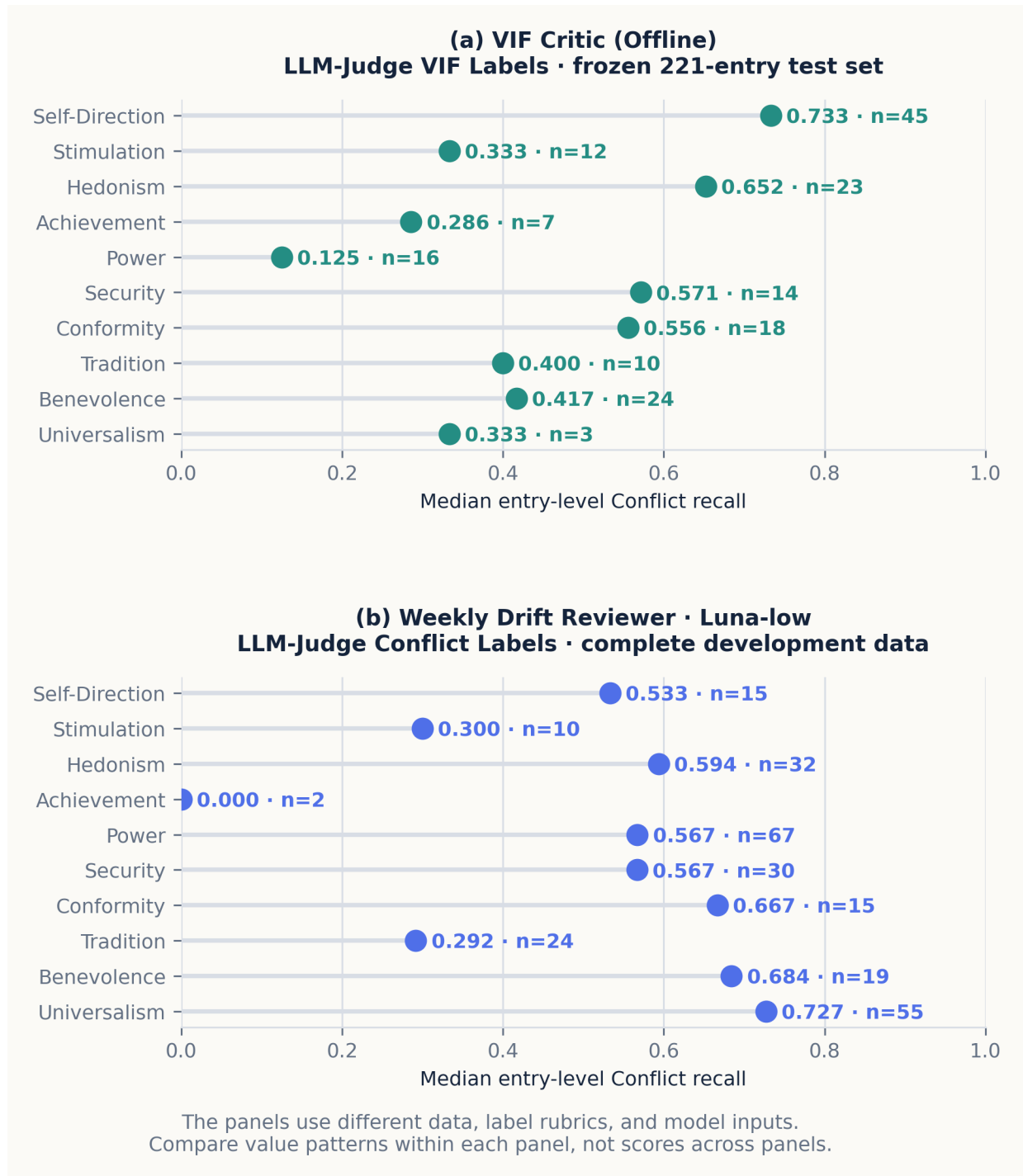


Figure 8: Per-value entry-level Conflict recall under the two completed but non-comparable evaluation regimes. Panel (a) uses LLM-Judge VIF Labels for the VIF Critic (Offline) on the frozen 221-Journal-Entry test set; panel (b) uses LLM-Judge Conflict Labels for Luna-low on complete development data. Each label gives median recall and Conflict support.

The research programme deliberately tested whether the VIF Critic (Offline) could meet the task before assigning a more capable model to the user-facing path. The resulting plateau is informative: the VIF Critic

(Offline) remains a small, reproducible Pattern Recognition Systems contribution, while its Conflict recall and value-specific instability are too weak for user-facing Drift authority.

Investigation 2 finding: The compact VIF Critic (Offline) captures reproducible ordinal and Conflict signal, but no tested intervention removes the long-tail, target, and context limits. Its median QWK and Conflict recall support an offline research contribution, not authority over user-facing Drift.

4.3 Investigation 3: the Weekly Drift Reviewer provides the stronger Weekly Drift Detection path

VIF Critic hand-off

Figure 9 shows the gpt-5.4-mini hand-off ablation. Without VIF Critic input, the Weekly Drift Reviewer found a median 9 of 33 Drifts. Raw VIF Critic Predictions reduced this result to 7 of 33 and added three median false Drift alerts, while VIF-Critic-triggered early calls plus Weekly Drift Detection retained 9 of 33 and added one median false Drift alert.

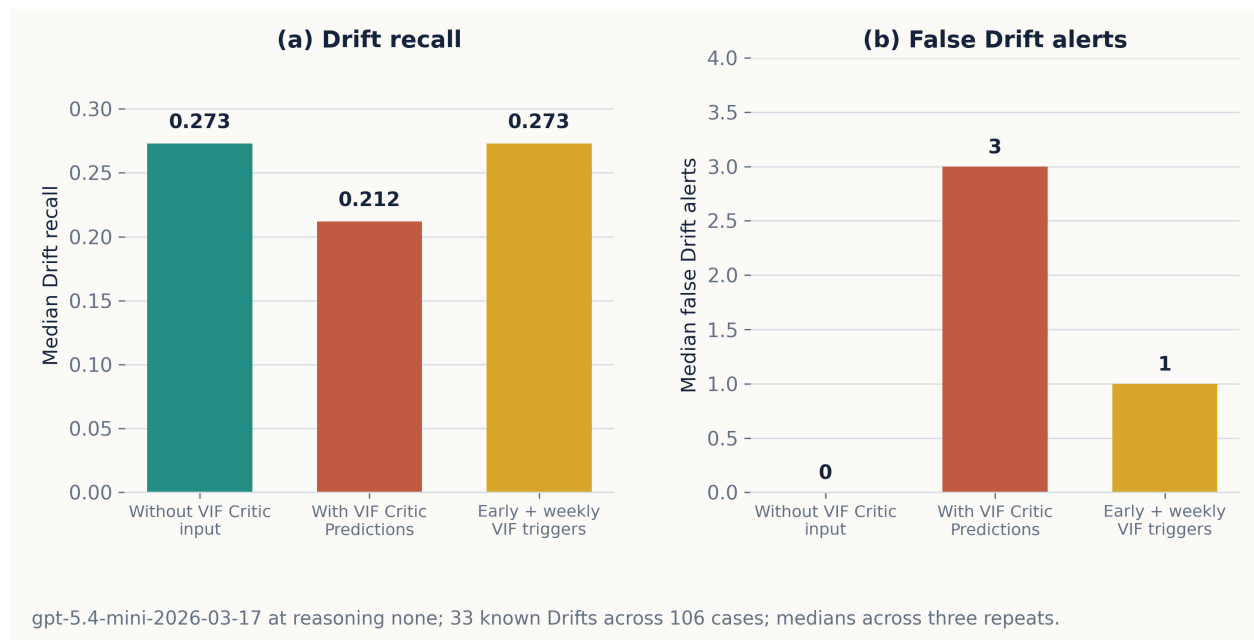


Figure 9: VIF Critic (Offline) hand-off ablation on the 33-Drift development union. All setups used gpt-5.4-mini-2026-03-17 at no reasoning effort for three repeats. Raw VIF Critic Predictions numerically lowered median Drift recall, but the paired interval included zero; early scheduling changed delay without increasing Drift hits.

The paired raw-input Drift-recall difference was -0.061 with a 95% interval from -0.158 to 0.033. The interval includes zero, so the recall loss is inconclusive. Coverage fell by 0.094 with a 95% interval from -0.170 to -0.019. VIF-Critic-triggered early calls reduced median delay from five days to one day, but the recall difference was exactly zero. The observed delay result came from development cases with historical training provenance and did not transfer to the non-training subgroup.

Hand-off finding: VIF Critic Predictions did not improve Drift recall in the tested gpt-5.4-mini input or scheduling ablations. No matched Luna-low VIF ablation was run.

Weekly Drift Reviewer model and reasoning-effort studies

The model comparison held the complete 292-case development data, prompt, reasoning effort, and three-repeat design constant. At no reasoning effort, gpt-5.4-mini-2026-03-17 found a median 7 of 42 known Drifts, giving Drift recall of 0.167 with five false Drift alerts. gpt-5.6-luna found 20 of 42, giving Drift recall of 0.476 with 13 false Drift alerts. The paired Drift-recall difference was +0.286 with a 95% interval from +0.158 to +0.425, while the false-alert difference was +9 with a 95% interval from +2 to +17. The stronger tested model therefore recovered substantially more known Drift, but it did so by accepting a more aggressive false-alert trade-off.

The later reasoning-effort study assessed direct Weekly Drift Reviewer operating points on the same complete 292-case, 42-Drift development data. These operating points are not comparable with the 106-case, 33-Drift hand-off union. Figure 10 shows the trade-off. Low reasoning effort had 0.548 median Drift recall, four median false Drift alerts across all 292 Persona/Core Value histories, 0.852 median Drift precision, and 0.637 trajectory coverage (a median 186 covered histories). The false-alert total includes unmatched predictions in histories with reference Drift; three false alerts occurred in the 256 non-Drift histories in each repeat. Medium had no clear recall gain over low. High reached 0.619 recall with eight false Drift alerts, but its paired recall interval against low included zero. Xhigh reached 0.667 recall with nine false Drift alerts and 0.750 median Drift precision. Against low, the xhigh paired Drift-recall difference was +0.095 with a 95% interval from +0.023 to +0.186, and the false-alert difference was +5 with a 95% interval from +1 to +9. Xhigh is therefore a more aggressive operating point, not a clean improvement.

Panel (b) of Figure 8 shows that Luna-low also had uneven entry-level Conflict recall. Median recall was 0.727 for Universalism, 0.684 for Benevolence, and 0.667 for Conformity, compared with 0.300 for Stimulation and 0.292 for Tradition. Achievement recall was zero, but only two reference Conflicts were available. These values describe error concentration within the Weekly Drift Reviewer study; they are not a direct comparison with the VIF Critic (Offline) panel because the data, reference labels, and model inputs differ.

Architecture decision

Low reasoning effort did not satisfy the original preregistered selection rule, which capped coverage loss at 0.05 and therefore mechanically retained no reasoning effort. After reviewing the development results, the project replaced that rule with the adopted hierarchy of Drift recall first, false Drift alerts second, and coverage as a diagnostic, then selected low reasoning effort. Its recall difference against no reasoning effort was +0.071 with a 95% interval from -0.071 to +0.205, while false Drift alerts fell by nine and coverage fell by 0.140 with intervals that excluded zero. The later xhigh result ranks higher on recall under that hierarchy, but the project retained low after declining an increase from four to nine median false Drift alerts accompanying a rise from 23 to 28 median Drift detections. The fixed contract is therefore a documented capstone choice rather than a preregistered optimum or deployment threshold, and it remains subject to a fresh final test.

Investigation 3 finding: Reviewing cumulative Journal Entry history with the Weekly Drift Reviewer is the stronger tested Weekly Drift Detection path. The fixed Luna-low contract is an evidence-informed capstone choice that balances Drift recall, false Drift alerts, and coverage; it is not a universal optimum, fresh final-test result, or deployment threshold. The VIF Critic (Offline) remains outside the user-facing path.

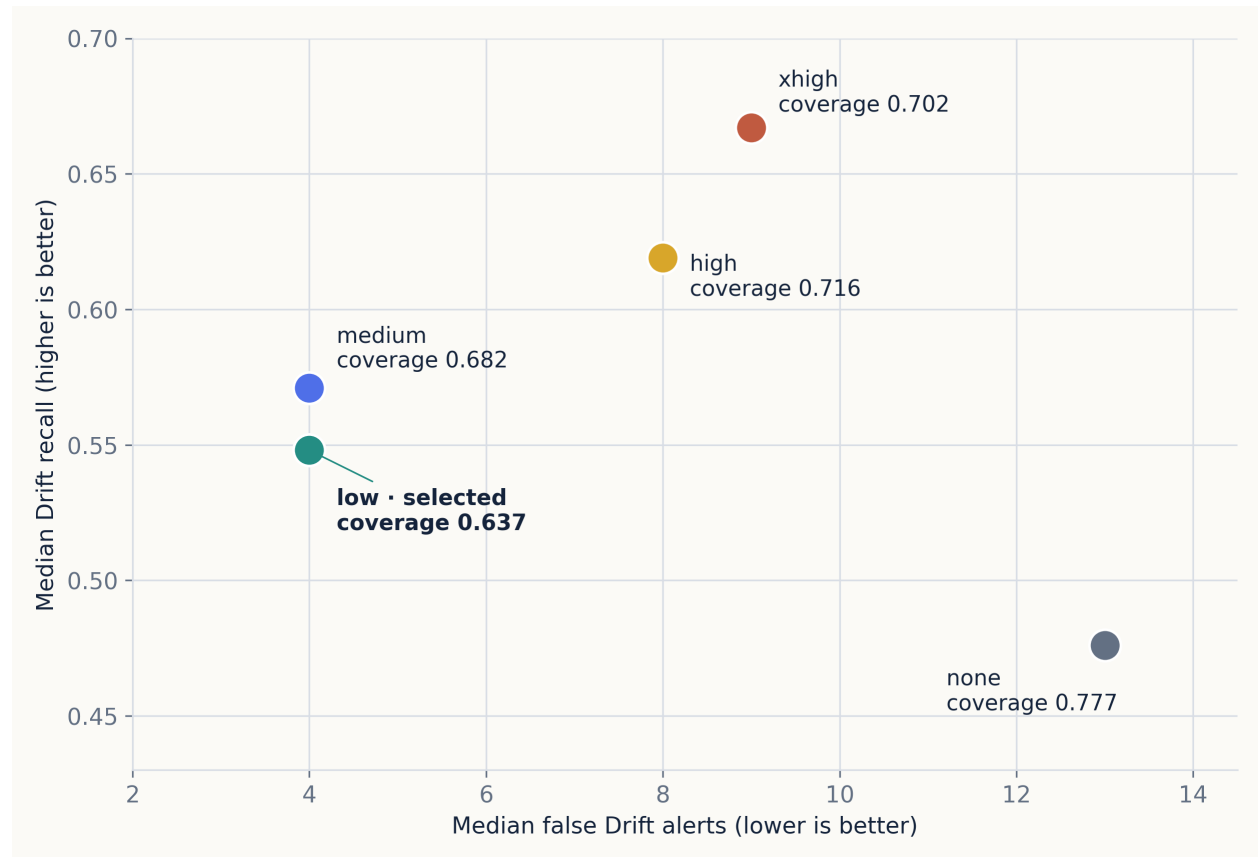


Figure 10: Weekly Drift Reviewer operating points on AI-reviewed synthetic development data. Each marker has an exact trajectory-coverage label; marker area carries no additional quantity. Twinkl retains low reasoning effort as the fixed capstone contract.

Effect of Core Value definitions on Drift detection

The [Core Value definition study](#) tested whether supplying definitions and motivational descriptions changed detected Drifts. These descriptions combine Schwartz-based definitions with project-specific elaborations. Both prompt setups used the same 204 Personas, 951 observed weeks, Journal Entry input, model settings, response format, and deterministic rules. Three repeats per prompt setup produced 5,706 requests, interleaved in a randomised order.

Adding the descriptions reduced median detected Drifts from 22 to 19; totals across the three repeats were 22, 22, and 21 without descriptions and 19, 21, and 18 with them. Between 305 and 319 of the 2,377 Journal Entry/Core Value decisions changed in each paired repeat. Aggregate counts also concealed changes in which Personas were identified: one repeat retained two Conformity Drifts while identifying different Personas in each prompt setup. The analysis retained 22 invalid reviews as Abstain decisions.

The comparison shows that additional value context can alter both detection frequency and the histories identified. It did not measure accuracy against reference labels, and variation between repeats limits attribution of individual changes. The accuracy figures reported earlier in this section describe the evaluated configuration and have not been re-established for the revised prompt.

4.4 Implementation objective: application verification and inspection

The React application connects the confirmed Profile, manual Journal Entry capture, displayed-nudge response, explicit closed-week review, Weekly Drift Detection, Coach Digest, and Inspect evidence through one versioned session. Figure 2 shows the user-facing selection step, while Figure 11 shows the resulting weekly reflection. Saved Persona replays allow direct selection of any week while restricting each selected result to evidence available at its cutoff. An assessor can move from the displayed result to the prompt, response, and source assessments illustrated in Figure 5 and Figure 4.

Nisha's confirmed Profile includes Universalism. In the saved fourth-week review, the Weekly Drift Reviewer identifies consecutive Conflicts on 8 and 9 March: remaining silent after a colleague's disparaging comment and choosing a chain store despite an intention to support a women's group. The Drift Detector confirms Active Drift on 9 March. The selected North Star Moment comes from 7 March, before Drift onset, when Nisha helped repair a classroom desk. As Table 7 shows, the earlier supportive action accompanies the reflection without cancelling the current Drift state. Other saved Personas demonstrate No Active Drift, continuing Drift, Insufficient Evidence, and different concurrent Core Value states.

Application stage	Saved evidence	Displayed result
Profile	Universalism is a confirmed Core Value	Universalism is the reference for later review
Journal Entries	8 March: remaining silent; 9 March: choosing the chain store instead of waiting to support the women's group	Two consecutive Conflicts with Universalism
Weekly Drift Detection	Drift onset 8 March; confirmation 9 March; current run length 2	Active Drift
Coach Digest	Connects the recent choices with the earlier classroom assistance	An evidence-based reflection and one question about the difference between those moments

Application stage	Saved evidence	Displayed result
North Star Moment	7 March: “Stayed back to help the peon uncle fix a broken desk in my classroom.”	An earlier supportive action, with links to its source and Inspect
Inspect	Recorded prompts, responses, source assessments, validation, and Drift Detector state	The assessor can trace the reflection and state to the available evidence

Table 7. Application walkthrough for Nisha’s saved fourth-week replay, with North Star Moment context selected.

Verification results and failure recovery

The [September verification assessment](#) passed the full Python suite, frontend suite, and two Chromium journeys shown in Table 8. These checks cover application behaviour and research reproducibility, with controlled model responses used for browser testing.

Verification	Result	Scope
Python	2,111 passed	Full repository suite, including research compatibility and application tests
Frontend	385 passed across 24 files	React component and integration tests
Browser	Two journeys passed	React application and Python service in Chromium at a mobile viewport
Build and lint	Passed	TypeScript/Vite build and Ruff for changed Python files
Type checking	Eight focused files passed	Existing errors remained in two additionally checked modules; not a clean repository-wide MyPy result

Table 8. Verification completed without paid model calls. Passing checks establish the exercised behaviour; they do not measure semantic model quality.

The browser journeys exercised saved replay through Inspect and reload, and manual onboarding through Journal Entry capture, closed-week review, Coach Digest failure and retry, and confirmed deletion. Trace assertions verified that retrying Coach Digest did not duplicate a Journal Entry or repeat the completed Weekly Drift Reviewer work, while deletion removed the corresponding Python session as well as resetting browser state. These checks test recovery and cross-component consistency in addition to the successful path.

Verification identified defects that affected both displayed responses and evaluation integrity. The grounding rule had accepted a response containing one genuine and one invented quotation; validation now checks every extracted quotation. Another defect discarded prior-week evidence during persistence and changed the assessment of a supported statement. Further corrections aligned retry behaviour with recorded model

Drift Detection (End of Week)

(based on 7 Journal Entries through Mar 9; includes earlier weeks)

[Read Journal Entries](#)

Active Drift

A repeated conflict with a Core Value is now active.

► WHY THIS STATE

Inspect decision

COACH DIGEST

Without North Star Moment

Showing: With North Star Moment

You stayed back to help the peon uncle fix a broken desk, while later you kept quiet with Sharma sir because you were tired and he was senior. Then, at the Hazratganj shop, you chose speed and spent more rather than waiting another day: "So I spent probably twice as much at a chain store when I could have waited one more day and supported the women's group instead."

Across these recent choices, tiredness, hierarchy, and convenience seem to have narrowed the space for the fairer or more considerate option you might otherwise have held open. That sits alongside the care you showed at school, making the week feel less like one simple story and more like a series of moments where your energy and circumstances mattered.

When you think about staying to fix the desk and later choosing the chain store, what feels most important about the difference between those two moments?

NORTH STAR MOMENT

A past moment in your own words

Making the world a fairer, better place

You've acted on what matters to you before. You wrote:

Stayed back to help the peon uncle fix a broken desk in my classroom.

[Open Journal Entry · Mar 7, 2025](#)

Inspect this moment

[Mar 2](#)[Mar 7](#)[Mar 8](#)[Mar 9](#)

Figure 11: Nisha's fourth-week Experience with the North Star Moment comparison selected. Active Drift remains visible alongside the Coach Digest, its reflective question, and an earlier supportive action linked to the original Journal Entry and Inspect. Saved synthetic replay.

attempts and prevented duplicate label records from concealing missing Journal Entries. Regression tests preserve these corrections.

Frontend inspection of prompt telemetry

A local browser walkthrough followed Nisha's fourth-week result into Inspect and examined the saved Coach Digest comparison and North Star Moment request. The displayed records allowed the assessor to check the supplied source text, distinguish initial instructions from repairs, and follow the recorded response through validation to the displayed result. Frontend tests also verified that the displayed prompt records matched the saved receipts and that a later retry preserved the original weekly context. These observations corroborate the inspection method described in Section 3.11; supporting code and test records are listed in Appendix A.

Coach Digest generation and semantic review

Across the five Persona key weeks deployed in August ($n = 5$), all accepted responses passed the Coach Digest Validations. These responses and scores retain their original input provenance; they do not establish the quality of responses to the revised replay inputs. The same-model Coach Digest Evals had mean scores of 4.80 for correctness, 5.00 for specificity, 5.00 for non-prescriptive tone, and 4.60 for tension honesty; all five reflective questions passed. Four responses scored 5 for tension honesty, while one scored 3 because the evaluator judged that it risked implying a current tension for two Core Values in a No Active Drift state. With five same-model reviews, this observation identifies a concrete target for future human calibration of the AI review rather than an error rate. These scores assess whether the saved responses follow the intended evidence and tone contracts. They do not measure sycophancy as a model property or establish human-perceived quality. Appendix B records the model-call, token, latency, and published-rate details for reproduction.

The [September voice refresh](#) and [paired source-context generation](#) extend the development record to the current roster. AI editorial review identified unsupported motives, incorrect chronology, and interpretations based on incomplete excerpts, even when automated checks passed. A [subsequent two-response pilot](#) also revealed that intended guidance had been omitted from the model's instructions. Correcting that omission and testing the assembled request addressed the implementation defect, while the interpretation errors required review against the original Journal Entries.

Reassessment under the strengthened validation rules passed all 44 comparison responses and 21 of the 27 base responses. Of the 27 Coach Digests displayed in the default replay, 25 passed. The two remaining responses contained source-punctuation mismatches, and one also failed a language check; the quotation findings did not establish fabricated content. This reassessment is separate from the August semantic scores.

The displayed nudge is implemented and appears in the application flow, but it has not been separately evaluated. The application also separates stable provider instructions from user-controlled JSON and saves a prompt-boundary receipt. Confirmed deletion clears browser state and the matching temporary Python session; the application does not claim deletion from the AI provider.

Implementation finding: Twinkl connects Profile construction, Weekly Drift Detection, and Coach Digest through a tested and inspectable application. Automated verification protects evidence and state consistency, while Inspect supports review of the prompts and responses behind each result. Source review remains necessary for interpretation errors that deterministic checks cannot resolve.

4.5 North Star Moment: results after the targeted update

Full eligible history achieved higher Card precision and Opportunity recall than Nomic top-three retrieval in both partitions after the targeted update. Table 9 reports the metric-specific numerators and denominators after the fixed shared-reference adjudication and paired exclusions. An incorrect displayed selection can contribute both a false positive for precision and a missed opportunity for recall; displaying a supportive quotation alone is insufficient if it violates full-history selection priority.

Partition	NSM method	Card precision	Opportunity recall
Development	Nomic top three	181/309 = 58.58%	181/327 = 55.35%
Development	Full eligible history	244/323 = 75.54%	244/327 = 74.62%
Qualified final	Nomic top three	49/85 = 57.65%	49/86 = 56.98%
Qualified final	Full eligible history	70/85 = 82.35%	70/86 = 81.40%

Table 9. AI-reviewed NSM results across 391 development and 110 qualified-final weeks.

The paired Nomic-minus-full-history comparison produced development differences of -16.97 percentage points for Card precision (95% interval -23.14 to -10.72) and -19.27 points for Opportunity recall (-25.45 to -12.72), and qualified-final differences of -24.71 percentage points for Card precision (95% interval -36.18 to -10.75) and -24.42 points for Opportunity recall (-36.47 to -9.88). These intervals favour full history under the tested configuration; they do not establish a general superiority claim across users or evaluators.

The targeted execution added 202 generation attempts with \$0.90767 in known API costs; five timed-out attempts lack usage receipts, so that cost is incomplete. Historical costs and retained latency observations are reported separately in the consolidated record. The unchanged consistency sample retained source-decision agreement of 341/384 (88.80%) and exact-quotation acceptance agreement of 107/109 (98.17%). These are AI-consistency measurements, not human validation, and the mixture of historical and new timing observations does not form an independent latency benchmark. The [complete report](#) and linked record preserve per-case differences, failed requests, exclusion lists, source lineage, costs, and all intervals.

5. Discussion

5.1 Main findings

The first investigation shows that synthetic-data construction is part of the research contribution rather than a preliminary engineering task. Sequential generation supplied earlier Journal Entries to support narrative continuity, while configured demographic and writing variation broadened designed coverage. Banned terms and the removal of generation metadata from production-like decisions reduced two direct leakage paths. Most importantly, the frozen targeted-batch loop converted model failures into testable data hypotheses. Power improved when the first batch targeted mild Power and Security Conflicts, but the absence of a family-wide gain and the later Security regression showed that additional synthetic data could move one boundary while worsening another.

The human-overlap benchmark strengthens but also bounds that conclusion. Aggregate agreement supports using the LLM-Judge VIF Labels for development, yet lower agreement on several values, project-team annotators, same-family generation and labelling, and missing original model provenance prevent a human-ground-truth claim. Designed diversity is similarly narrower than realism: varying age, profession, culture, tone, and chronology does not show that real users would write or be interpreted in the same way.

The second investigation shows why the project began with a deliberately compact model. The VIF Critic (Offline) was sufficient to test whether frozen embeddings and Profile weights could recover ordinal value alignment from one Journal Entry. Its 133 persisted configurations tested several possible explanations for the plateau, including loss choice, encoder choice, checkpoint retention, small train-only data lifts, and compact history. The negative result is therefore informative within the tested design space: the MLP learned signal, but not enough stable Conflict signal to receive user-facing Drift authority.

The third investigation then answered the architectural question. VIF Critic Predictions did not improve Drift recall in the tested gpt-5.4-mini input or scheduling setups, and the early-call delay benefit did not extend to the non-training subgroup. The model comparison found more known Drifts with Luna than with gpt-5.4-mini under the same cumulative-history input contract, and later reasoning-effort experiments exposed the recall, false-alert, and coverage frontier used to retain Luna-low. This progression is evidence-driven component selection rather than an apples-to-apples claim that an LLM outscored the MLP.

The resulting architecture preserves all three contributions. The corpus and label studies establish the development evidence; the VIF Critic (Offline) records the Pattern Recognition Systems investigation and its negative frontier; and the Weekly Drift Reviewer with the deterministic Drift Detector owns user-facing Drift. Inspect exposes the model receipts, validations, and rule transitions. The selected Weekly Drift Reviewer is not a universal optimum: higher reasoning effort raises recall and false Drift alerts together, while low reasoning effort abstains more than no reasoning effort. Accordingly, a failed review yields Insufficient Evidence; an Abstain or Journal Entry gap does so only when it blocks a current Drift claim after recent Conflict evidence. A standalone valid Abstain with no recent Conflict evidence remains No Active Drift.

Across the investigations, experimental findings informed component selection and subsequent verification. Table 10 summarises how local improvements, negative results, and performance trade-offs shaped the implemented application. Inspect and regression tests support review of those decisions by preserving the evidence and checking the resulting behaviour.

Development question	Controlled comparison or verification	Finding and resulting decision
Does targeted synthetic data improve weak values?	Train-only augmentation with frozen validation/test Personas and matched training seeds	Local gains coexisted with regressions; retain the historical reference rather than infer a general gain from corpus growth.
Does the VIF Critic (Offline) improve downstream Drift detection?	Fixed gpt-5.4-mini input and scheduling ablations	No Drift-recall gain in the tested setups; keep VIF Critic Predictions outside the adopted user-facing path.
Which reasoning effort suits the capstone?	Same development references across model settings and repeated Runs	Greater recall can add false alerts; retain Luna-low as a documented trade-off.

Development question	Controlled comparison or verification	Finding and resulting decision
Do selected Core Value definitions change decisions?	Matched prompts with and without definitions, using identical histories and rules	Detection counts and identified histories changed; accuracy of the revised prompt remains to be evaluated.
Does retrieval preserve NSM selection quality?	Paired Nomic top-three and full-history review with shared AI references and whole-Persona intervals	Full history had higher Card precision and Opportunity recall; adopt it for the tested capstone configuration.
Can the application preserve evidence and recover from failure?	Data-integrity, model-call, persistence, and browser tests	Corrected defects are covered by regression tests; source interpretation still requires semantic review.

Table 10. Relationship between development questions, findings, and implementation decisions.

5.2 Cross-value findings and next analysis

The aligned panels in Figure 8 make one useful descriptive comparison possible without claiming score equivalence. Power had the lowest VIF Critic (Offline) Conflict recall at 0.125 but reached 0.567 under Luna-low; Universalism moved from 0.333 on only three VIF test Conflicts to 0.727 on 55 complete-development Conflicts. In contrast, Stimulation remained weak in both panels at 0.333 and 0.300, while Tradition was 0.400 and 0.292. Self-Direction moved in the opposite direction, from 0.733 to 0.533. These shifts identify values for deeper analysis, but they may reflect different label definitions, supports, Core Value selection, visible history, or data composition rather than model capability.

A matched component-level error analysis can use the existing frozen 106-case hand-off union. That stored evidence contains VIF Critic Predictions, resolved LLM-Judge Conflict Labels, and Weekly Drift Reviewer Decisions for the same Journal Entry/Core Value combinations. A matched analysis can report per-value recall, precision, abstention, support, and error overlap, then examine exact Journal Entries where both components fail, where the Weekly Drift Reviewer corrects a VIF Critic error, and where prior entries may explain a difference in interpretation. Because those matched per-value metrics and qualitative cases have not yet been produced, this paper treats them as the next analysis rather than presenting the current non-comparable panels as a causal result.

5.3 Validity limits

The largest limitation is the synthetic corpus. Demographic and narrative controls establish designed coverage rather than population representativeness, while the original Claude Code generation and label files lack a stable model identifier. Because Claude Code subagents produced both the Journal Entries and their original LLM-Judge VIF Labels, generator and judge errors may be correlated; the study did not measure that correlation.

The blind-first human benchmark contains only 115 Journal Entries from 19 personas, and its annotators—Des, JL, and KM—belong to the project team rather than an independent external panel. Stimulation has only two Core Value personas in the shared sample, and kappa depends on category prevalence as well as observed agreement [24]. The aggregate agreement statistics must therefore remain paired with the per-dimension result and cannot define a human-consistency ceiling.

The VIF Critic (Offline) studies retained the same validation and test personas across targeted augmentation, preventing partition overlap and changes in holdout composition. However, the test results informed later augmentation and reference-model decisions. The 221-Journal-Entry test partition therefore became a repeatedly consulted historical research benchmark rather than an untouched final test of the full development programme.

All Weekly Drift Detection references are AI-reviewed synthetic development evidence, and some cases have historical training provenance. The 42-Drift complete-development study was used to choose the fixed model contract, so presenting the same results as fresh final-test performance would constitute leakage. Luna-low also failed its original preregistered coverage gate before the project replaced that rule with the later development-selection hierarchy, and the subsequent decision to retain low despite the xhigh recall gain is an explicit capstone trade-off. The earlier VIF hand-off study used gpt-5.4-mini; without a matched Luna-low VIF ablation, its result cannot determine whether VIF Critic input would alter the adopted Luna-low operating point. The trajectory bootstrap also resamples Core Value histories separately even when they share one Persona’s writing and model calls; the robustness of the intervals to resampling whole Personas has not been tested.

Application evidence has similar limits. The five August Coach Digest Evals are too few to establish content quality, and the same model generated and evaluated those responses. Later generations and AI editorial review broaden the development record without expanding that scored sample or establishing human-perceived usefulness. The behavioural corpus and displayed nudge still require semantic evaluation. Browser testing covers selected recovery paths in Chromium, while existing type-checking errors and two validation failures among displayed Coach Digests remain unresolved. Further testing and independent review are therefore needed before claims about operational reliability or deployment readiness.

The NSM comparison assesses synthetic writing with an AI evaluator from the same model family as the runtime reviewer. Its qualified-final histories were separated from recorded NSM semantic development, but had prior upstream research exposure, including use in the broader Weekly Drift development comparison. Retaining 463 prior observations in the targeted update also limits its interpretation as independent replication. The results therefore support method selection for the tested configuration, while human assessment of quotation suitability and real-user usefulness remain open.

5.4 Safety, privacy, and ethics

Journal Entries can contain sensitive personal information. The application therefore gives a first-use notice for browser storage, temporary Python memory, provider use, assessment-only scope, and the non-therapy boundary; invalid model output fails closed, and Inspect shows saved model evidence. These controls support informed inspection but do not replace a privacy review or security assessment.

The application must avoid moralising a person’s values. A Conflict is a behaviour-level decision against one declared Core Value in the available text, not a judgement of character, and Drift requires repeated Conflict evidence. Coach Digest therefore uses open questions and does not prescribe action.

Users can expect a level of understanding beyond what conversational agents can provide [9]. Twinkl seeks to address this risk with cited Journal Entries, Insufficient Evidence, and explicit AI-review labels. A real-user pilot must still test whether this language works in practice.

6. Conclusions and Future Work

Twinkl investigates a narrower role for AI-assisted reflection: longitudinal accountability to the user's own Core Values. Schwartz's Theory of Basic Human Values supplies the theoretical foundation, while cited Journal Entries, Insufficient Evidence, and a non-prescriptive Coach Digest are intended to reduce the risk of ungrounded affirmation. The project does not measure sycophancy or establish behaviour change, but it turns accountability into a concrete Intelligent Systems problem that can be modelled, tested, and inspected.

The three investigations establish a connected research path. First, the synthetic corpus is suitable for bounded model development because its generation, train-only augmentation, labelling, and agreement evidence are explicit and reviewable; it is not a substitute for real-user data or independent human labels. Second, the compact VIF Critic (Offline) recovers measurable ordinal and Conflict signal, but its 69-run-ID, 133-configuration programme reaches an inadequate and value-dependent frontier. Third, VIF Critic Predictions did not improve Drift recall in the tested Weekly Drift Detection hand-offs, whereas the Weekly Drift Reviewer supports the adopted longitudinal path with an explicit recall, false-alert, and coverage trade-off.

Twinkl therefore retains the VIF Critic (Offline) as a Pattern Recognition Systems contribution and gives user-facing Drift authority to the Weekly Drift Reviewer followed by the deterministic Drift Detector. The implementation integrates these responsibilities through tested evidence preservation and failure recovery. Inspect makes the resulting decisions reviewable by exposing the supplied prompts and evidence alongside recorded responses, validation, and Drift state transitions.

The current evidence supports bounded use of the LLM-Judge VIF Labels for development and an implemented and inspectable core assessment path. It does not support a direct MLP-versus-LLM performance claim, real-user benefit, or deployment readiness. The Technical Paper contributes the methodology, experiments, negative results, and architecture decision; the public assessment application and saved replays separately support the System Implementation & Demo assessment.

Future work should begin with the stored matched per-value error analysis described in Section 5.2, including exact Journal Entry case studies, and semantic review of the behavioural regression corpus and combined Coach Digest/NSM responses. A frozen final test must exclude model and prompt development data and evaluate the exact adopted prompt contract. Future human calibration of the AI review with independent reviewers and a five-to-ten-user pilot should then examine perceived Coach Digest accuracy, relevance, timing, displayed-nudge response, and continued journaling over one to two weeks. A matched Luna-low VIF ablation would isolate whether the offline signal has value under the adopted reviewer contract, while provider attack testing, privacy review, and controlled latency measurement remain necessary before any deployment claim.

The [NSM evaluation](#) now supports full eligible history over the tested Nomic shortlist for Card precision and Opportunity recall. Saved replay connects the completed full-history outcomes to the five selected Personas while preserving exact quotations and receipt provenance. The final source-level walkthrough remains separate work, and live execution requires a valid budget; a human study is still required to assess usefulness.

AI Tool Declaration

We used OpenAI Codex to inspect repository evidence, create report figures, capture local application screenshots, examine Inspect in the browser, and help draft and edit this paper. Codex also assisted with implementation and regression tests and authored the labelled synthetic behavioural examples identified in Section 3.8. We checked numerical claims against the named evaluation reports, current code, and stored run records. The authors remain responsible for study design, interpretation, source verification, and submitted text.

The project also used language models for synthetic Journal Entry generation, LLM-Judge VIF Labels, LLM-Judge Conflict Labels, Weekly Drift Reviewer Decisions, Coach Digest generation, Coach Digest Evals, and NSM quotation selection. Each use and its evidence limit are stated in the relevant method or result section.

References

- [1] I. Li, A. K. Dey, and J. Forlizzi, “A stage-based model of personal informatics systems,” in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pp. 557–566, 2010. <https://doi.org/10.1145/1753326.1753409>
- [2] D. A. Epstein, A. Ping, J. Fogarty, and S. A. Munson, “A lived informatics model of personal informatics,” in *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, pp. 731–742, 2015. <https://doi.org/10.1145/2750858.2804250>
- [3] E. P. S. Baumer, “Reflective informatics: Conceptual dimensions for designing technologies of reflection,” in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, pp. 585–594, 2015. <https://doi.org/10.1145/2702123.2702234>
- [4] M. Sharma *et al.*, “Towards understanding sycophancy in language models,” in *The Twelfth International Conference on Learning Representations*, 2024. https://proceedings.iclr.cc/paper_files/paper/2024/file/0105f7972202c1d4fb817da9f21a9663-Paper-Conference.pdf
- [5] S. H. Schwartz, “Universals in the content and structure of values: Theoretical advances and empirical tests in 20 countries,” *Advances in Experimental Social Psychology*, vol. 25, pp. 1–65, 1992. [https://doi.org/10.1016/S0065-2601\(08\)60281-6](https://doi.org/10.1016/S0065-2601(08)60281-6)
- [6] S. H. Schwartz *et al.*, “Refining the theory of basic individual values,” *Journal of Personality and Social Psychology*, vol. 103, no. 4, pp. 663–688, 2012. <https://doi.org/10.1037/a0029393>
- [7] J. A. Lee, G. N. Soutar, and J. J. Louviere, “The best-worst scaling approach: An alternative to Schwartz’s values survey,” *Journal of Personality Assessment*, vol. 90, no. 4, pp. 335–347, 2008. <https://doi.org/10.1080/00223890802107925>
- [8] A. A. J. Marley and J. J. Louviere, “Some probabilistic models of best, worst, and best-worst choices,” *Journal of Mathematical Psychology*, vol. 49, no. 6, pp. 464–480, 2005. <https://doi.org/10.1016/j.jmp.2005.05.003>
- [9] H. Gaffney, W. Mansell, and S. Tai, “Conversational agents in the treatment of mental health problems: Mixed-method systematic review,” *JMIR Mental Health*, vol. 6, no. 10, e14166, 2019. <https://doi.org/10.2196/14166>

- [10] A. Wang, *Master of Technology in Intelligent Systems Capstone Project: Capstone Requirements (ISS 2025–2026)*. Institute of Systems Science, National University of Singapore, 2025, slide 5. [Stable repository copy](#)
- [11] S. Nepal *et al.*, “MindScape Study: Integrating LLM and behavioral sensing for personalized AI-driven journaling experiences,” *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, no. 4, Article 186, 2024. <https://doi.org/10.1145/3699761>
- [12] S. Wu, Y. R. Fung, C. Qian, J. Kim, D. Hakkani-Tur, and H. Ji, “Aligning LLMs with individual preferences via interaction,” in *Proceedings of the 31st International Conference on Computational Linguistics*, pp. 7648–7662, 2025. <https://aclanthology.org/2025.coling-main.511/>
- [13] Z. Tan *et al.*, “In prospect and retrospect: Reflective memory management for long-term personalized dialogue agents,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8416–8439, 2025. <https://doi.org/10.18653/v1/2025.acl-long.413>
- [14] X. He, I. Nassar, J. Kiros, G. Haffari, and M. Norouzi, “Generate, annotate, and learn: NLP with synthetic text,” *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 826–842, 2022. https://doi.org/10.1162/tacl_a_00492
- [15] L. Zheng *et al.*, “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena,” in *Advances in Neural Information Processing Systems 36*, Datasets and Benchmarks Track, pp. 46595–46623, 2023. <https://doi.org/10.52202/075280-2020>
- [16] E. Tabassi, *Artificial Intelligence Risk Management Framework (AI RMF 1.0)*, NIST AI 100-1. Gaithersburg, MD: National Institute of Standards and Technology, 2023. <https://doi.org/10.6028/NIST.AI.100-1>
- [17] J. Cohen, “Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit,” *Psychological Bulletin*, vol. 70, no. 4, pp. 213–220, 1968. <https://doi.org/10.1037/h0026256>
- [18] W. Cao, V. Mirjalili, and S. Raschka, “Rank consistent ordinal regression for neural networks with application to age estimation,” *Pattern Recognition Letters*, vol. 140, pp. 325–331, 2020. <https://doi.org/10.1016/j.patrec.2020.11.008>
- [19] J. Ren *et al.*, “Balanced Meta-Softmax for long-tailed visual recognition,” in *Advances in Neural Information Processing Systems 33*, pp. 4175–4186, 2020. <https://proceedings.neurips.cc/paper/2020/hash/2ba61cc3a8f44143e1f2f13b2b729ab3-Abstract.html>
- [20] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in *Proceedings of the 33rd International Conference on Machine Learning*, pp. 1050–1059, 2016. <https://proceedings.mlr.press/v48/gal16.html>
- [21] Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar, “Nomic Embed: Training a reproducible long context text embedder,” arXiv:2402.01613, 2024. <https://arxiv.org/abs/2402.01613>
- [22] J. L. Fleiss, “Measuring nominal scale agreement among many raters,” *Psychological Bulletin*, vol. 76, no. 5, pp. 378–382, 1971. <https://doi.org/10.1037/h0031619>
- [23] J. Cohen, “A coefficient of agreement for nominal scales,” *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960. <https://doi.org/10.1177/001316446002000104>
- [24] T. Byrt, J. Bishop, and J. B. Carlin, “Bias, prevalence and kappa,” *Journal of Clinical Epidemiology*, vol. 46, no. 5, pp. 423–429, 1993. [https://doi.org/10.1016/0895-4356\(93\)90018-V](https://doi.org/10.1016/0895-4356(93)90018-V)

[25] OpenAI, “GPT-5.6 Luna,” *OpenAI API documentation*, accessed 31 August 2026. <https://developers.openai.com/api/docs/models/gpt-5.6-luna>

Appendix A. Reproduction and Evidence Map

The core historical experiment evidence snapshot used for this paper is commit `dd4bfa9d`. Commit `44ebcf25` preserves the earlier application walkthrough and 30 August verification. Current implementation, prompt telemetry, later evaluation records, and the 12 September test checkpoint use commit `3dbb3e43`. Table A1 keeps these snapshots distinct so that later implementation changes do not silently alter the provenance of historical findings.

The linked study records preserve the inputs, model settings, responses, and evaluation procedures for the Core Value definition study, NSM targeted update, and later Coach Digest generations. Appendix C records the scope of completed browser checks and the pending final walkthrough.

Claim	Stable evidence
Product intent and scope	<code>docs/prd.md</code>
Profile construction	<code>docs/onboarding/onboarding_spec.md</code>
Synthetic generation and targeted-batch controls	<code>docs/pipeline/pipeline_specs.md</code>
Power/Security targeted data lift	<code>twinkl-681.5 review</code>
Hedonism/Security targeted data lift	<code>twinkl-691.3 review</code>
Human and LLM-Judge agreement on the shared 115-Journal-Entry subset	<code>docs/evals/judge_validation_summary.md</code>
Persisted LLM-Judge VIF Label distribution	<code>logs/judge_labels/judge_labels.parquet</code>
VIF Critic (Offline) evaluation	<code>docs/evals/value_modeling_eval.md</code>
VIF aggregate metric definitions	<code>src/vif/eval.py</code>
Balanced Softmax per-value test inputs	<code>run_019; run_020; run_021</code>
Historical trajectory analysis	<code>docs/drift/trajectory_eda.md</code>
VIF hand-off ablation	<code>twinkl-752.5 reassessment</code>
Complete Drift references	<code>twinkl-qtwz review</code>
Weekly Drift Reviewer model comparison	<code>twinkl-52zz model comparison</code>
Fixed low-reasoning comparison	<code>twinkl-52zz review</code>
Drift matching, coverage, and paired differences	<code>src/drift_rules.py; coverage definition; scoring and paired comparisons</code>
Luna-low per-value metrics	<code>metrics.json</code>
Higher-reasoning comparison	<code>twinkl-ck3w review</code>
Coach Digest sample and review	<code>docs/evals/overview.md</code>
Coach Digest operational measurements	<code>docs/evals/explanation_quality_eval.md</code>
Active Drift application walkthrough	<code>frontend/onboarding/public/scenarios/active-nisha.json</code>
Experience service and Python verification	<code>src/demo/experience_service.py; tests/demo/test_experience_service.py</code>
React Experience, saved Persona replay, and Inspect verification	<code>JournalExperience.tsx; PersonaReplay.tsx; InspectView.tsx; corresponding React tests</code>
Public assessment scope and URL	<code>docs/demo/experience_inspect_app.md</code>

Claim	Stable evidence
Prompt-boundary verification	Live prompt-boundary verification
Current testing and guardrail checkpoint	12 September assessment; saved-response reassessment
Current Inspect prompt telemetry	Frontend implementation; receipt projection tests; message-boundary renderer; live event construction
Browser and SDK failure verification	Chromium journeys; SDK transport tests
Controlled Weekly Drift Reviewer prompt comparison	Core Value definition study; accuracy evaluation scope
NSM paired methodology and targeted results	Methodology; targeted-update report; versioned record
Current Coach Digest generation and editorial review	27-response refresh; 22-pair comparison; selected voice pilot
Behavioural evaluation resource and historical replay	Eight-pair corpus; recorded-source replay helper

Table A1. Stable repository evidence for the paper's principal factual and methodological claims.

The following commands reproduce or re-score stored metrics, regenerate figures, and run application checks without paid model calls. Use the linked source snapshot appropriate to each claim and activate the repository virtual environment before Python commands. The first calculation restricts every Cohen comparison to the same 115-Journal-Entry intersection used for Fleiss' kappa. Run the score command in a clean checkout because it rewrites its tracked `metrics.json`, including the `scored_at` timestamp.

```
source .venv/bin/activate
uv run python - <<'PY'
from statistics import mean
import polars as pl
from src.annotation_tool.agreement_metrics import (
    calculate_cohen_kappa,
    calculate_fleiss_kappa,
    load_all_annotator_dfs,
    load_judge_labels,
)
from src.models.judge import SCHWARTZ_VALUE_ORDER

frames = load_all_annotator_dfs()
shared = set.intersection(*[
    set(frame.select("persona_id", "t_index").iter_rows())
    for frame in frames.values()
])
keys = pl.DataFrame(sorted(shared), schema={
    "persona_id": pl.String, "t_index": pl.Int64,
}, orient="row")
frames = {name: frame.join(keys, on=["persona_id", "t_index"])}
```

```
        for name, frame in frames.items()
fleiss = calculate_fleiss_kappa(list(frames.values()))
cohen = [calculate_cohen_kappa(frame, load_judge_labels())
        for frame in frames.values()]
print(f"shared_entries={len(shared)}")
for value in [*SCHWARTZ_VALUE_ORDER, "aggregate"]:
    mean_cohen = mean(result[value] for result in cohen)
    print(f"{value}: fleiss={fleiss[value]:.3f}, mean_cohen={mean_cohen:.3f}")
PY
uv run python -c \
    "import polars as pl; \
d = pl.read_parquet('logs/judge_labels/judge_labels.parquet'); \
print(d.select(pl.col('alignment_vector').explode().alias('label')) \
.group_by('label').len().sort('label'))"
uv run python -m \
    scripts.experiments.compare_twinkl_ck3w_luna_higher_reasoning score
MPLCONFIGDIR=/tmp/twinkl-matplotlib \
    uv run python scripts/capstone/generate_report_figures.py
uv run --no-sync python -m pytest
npm --prefix frontend/onboarding test
npm --prefix frontend/onboarding run build
npm --prefix frontend/onboarding exec -- playwright install chromium
npm --prefix frontend/onboarding run test:e2e
```

The full Python suite requires Git history containing the recorded study revisions. The Chromium install is required once per Playwright browser version; the browser command builds the frontend and starts the controlled Python API on local port 8765, which must be free. These checks do not execute the semantic behavioural corpus through a paid evaluator.

Paid model execution is not required to inspect the stored responses, reference labels, metrics, report figures, or saved Persona replays. The public Railway assessment is optional and may incur provider calls when used beyond the saved replays.

The application-verification environment on 30 August 2026 used Python 3.12.11 with uv 0.11.29, Node.js 22.17.1 with npm 10.9.2, and Quarto 1.10.18. Python dependencies are locked in `uv.lock`, while React dependencies are locked in `frontend/onboarding/package-lock.json`.

Appendix B. Coach Digest Operational Note

The [September base-response refresh](#) and its repairs used 40 provider calls, 74,171 input tokens, and 6,743 output tokens, with a calculated cost of US\$0.02004253. Separate request and cost records accompany the [source-context comparison](#) and [selected voice pilot](#). These later activities involved generation and AI editorial review without new Coach Digest Evals scores.

An [earlier September replay refresh](#) generated five accepted key-week responses using gpt-5.6-luna at no reasoning effort. Seven calls were required, including two validation-guided retries. The saved receipts

record 10,980 input tokens, 1,335 output tokens, and 25.383 seconds of summed request latency, with a calculated cost of US\$0.00434595. This refresh included no new semantic evaluation.

The five accepted August Persona key-week responses required seven generation calls because two initial responses failed Coach Digest Validations and were retried. Generation and Coach Digest Evals together used 12 calls, 16,547 input tokens, and 1,696 output tokens, with approximately 33.7 seconds of summed recorded request latency. Applying the provider's published standard-tier rates, including the cache-write multiplier, produced a total below one cent [25]. This is a reproduction calculation from saved receipts, not a billing record or a controlled latency benchmark.

Appendix C. NSM Saved Replay and Walkthrough Status

The [NSM evaluation and targeted update](#) are complete. Saved replay uses full-history runtime outcomes for Nisha, Noor, Lukas, Wei Jun, and Meera across 27 reviewed weeks: 22 selected quotations, three ineligible outcomes, and two completed reviews with no supportive source. Original provider receipts and source-availability conventions remain inspectable. The roster was selected for current Drift-state coverage and coherent source histories, not AI evaluation grades. The final source-level capstone walkthrough remains separate work, and live execution requires a valid separately configured budget. The August Coach Digest scores in Section 4.4 retain their historical scope.

A local browser check on 7 September 2026 verified the then-current Persona picker, Nisha's Active Drift key week with its NSM quotation and source drawer, and Sook Yin's ended Historical Drift Record with a No Active Drift card. Automated React projection checks covered that roster's 27 reviewed weeks. These observations precede the replacement of Sook Yin and Henrik with Lukas and Meera on 8 September and therefore do not establish browser verification of the new chooser or replacement replays. They remain evidence of the exercised behaviour, rather than an all-Persona browser walkthrough, human evaluation, or the final professor assessment.

Subsequent browser checks examined Lukas's paired Coach Digest responses and Nisha's fourth-week prompt records in Inspect. The report's current interface captures were taken on 12 September from Nisha's saved fourth-week replay, using the existing responses without new model calls. The two automated Chromium journeys reported in Section 4.4 additionally verified saved replay, reload, and the manual review and recovery path. These checks cover selected application behaviour; the final walkthrough across all five Personas remains pending.